



Knowledge-assisted hybrid optimization strategy of large-scale crude oil scheduling integrated production planning

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ABSTRACT

As modern refinery crude oil scheduling scales up, traditional manual operations and independent optimization methods struggle with complexity and dynamics. This study proposes an empirically assisted integrated planning and scheduling optimization model. The integrated model is solved using a hybrid optimization algorithm combining mathematical programming and particle swarm optimization (MP/PSO). Long-term planning aims to minimize operational and transportation costs while maximizing refinery profits; short-term scheduling, based on initial long-term plans, aims to minimize unit switchovers. In the short-term scheduling phase, heuristic rules based on empirical operational knowledge generate a high-performing initial population to accelerate convergence. This strategy is crucial for enhancing refinery emergency response capabilities, ensuring stable operations, and improving economic benefits. Experimental results show that within a reasonable time frame, MP/PSO performs better than PSO and manual scheduling in large-scale crude oil scheduling scenarios.

1. Introduction

1.1. Background

Crude oil scheduling optimization has been extensively researched and applied in refineries to achieve greater profits (Li et al., 2012). Crude oil scheduling is a crucial front-end process in refineries, playing an indispensable role in ensuring stable continuous production and increasing refining profits (Jia et al., 2003). The process mainly includes crude oil arrival, unloading operations, pipeline transportation, storage in plant tanks, and feeding into atmospheric and vacuum distillation units (Yang et al., 2023). Currently, crude oil scheduling primarily relies on manual experience and operations (Li et al., 2020), making it difficult to achieve integrated optimization from a global perspective to address fluctuations in the scheduling process. This approach cannot effectively and promptly respond to various uncertainties, leading to increased inventory, inefficient use of capital, and higher operating costs.

Existing scheduling optimization methods usually handle long-term planning or short-term scheduling separately (Wu et al., 2016). Long-term planning focuses primarily on maximizing refining profits and determining the types and quantities of crude oil, often overlooking the dynamic operational constraints in short-term scheduling. Although

short-term scheduling optimization can effectively address immediate operational constraints and efficiency issues, it lacks consideration of long-term economic and strategic goals. This separated optimization approach results in incomplete decision-making, failing to achieve comprehensive optimization and to fully utilize the advantages of long-term strategic goals and short-term operational flexibility.

Therefore, there is an urgent need for an integrated optimization method that combines planning and scheduling, leveraging the strengths of both long-term and short-term optimization. Such an integrated approach can not only enhance the overall efficiency and responsiveness of crude oil scheduling but also ensure consistency between operational decisions and strategic economic goals. By comprehensively utilizing insights from long-term planning and the flexibility of short-term operations, refineries can achieve more balanced and optimized scheduling strategies, thereby reducing operating costs, improving resource utilization efficiency, and enhancing economic benefits. This integrated approach is particularly significant in improving the emergency response capability of refineries, ensuring stable operations, and promoting the efficient management of petroleum resources.

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Notation			
Acronyms		Binary variables	
MP	mathematical programming	<i>Oilchange</i>	=1, indicates a change in the crude oil blend formula, otherwise 0.
PSO	particle swarm optimization	<i>Tankchange</i>	=1, indicates a change in the tank, otherwise 0.
Manual/PSO	Using the long-term plan from manual scheduling as the initial condition for PSO to solve the short-term scheduling.	<i>CTC_{m,t}</i>	=1, if there is a tank change supplying feed to CDU <i>m</i> during period <i>t</i> , otherwise 0.
MILP	mixed-integer linear programming	<i>XTankIn_{v,k,t}</i>	=1, indicates that vessel <i>v</i> unloads to tank <i>k</i> at period <i>t</i> , otherwise 0.
MILNP	Mixed-Integer Nonlinear Programming	<i>XTankOut_{k,m,t}</i>	=1, indicates that tank <i>k</i> feed to CDU <i>m</i> at period <i>t</i> , otherwise 0.
CDU	Crude Distillation Unit	<i>XT_{j,m,t}</i>	=1, indicates that virtual tank for crude <i>j</i> feed to CDU <i>m</i> at period <i>t</i> , otherwise 0.
Sets/Indices		<i>XC_{m,n,t}</i>	=1, indicates that CDU <i>m</i> product residue <i>n</i> at period <i>t</i> , otherwise 0.
<i>M</i>	set of the CDUs indexed by <i>m</i>	<i>Arrive_v</i>	=1, indicates that vessel <i>v</i> arrive period.
<i>T</i>	set of periods indexed by <i>t</i>	<i>XInxEmpty_{k,Arrive_v}</i>	=1, indicates that tank <i>k</i> is empty when being unloaded by vessel <i>v</i> , otherwise 0.
<i>V</i>	set of vessel indexed by <i>v</i>	Continuous variables	
<i>K</i>	set of tank indexed by <i>k</i>	<i>VD_{m,t}</i>	Total Feed of CDU <i>m</i> in period <i>t</i>
<i>J</i>	set of crude type indexed by <i>j</i>	<i>T_{A,v}</i>	The arrival time of the vessel <i>v</i> at the unloading location.
<i>P</i>	set of crude properties indexed by <i>p</i>	<i>T_{F,v}</i>	The completion time of unloading for the vessel <i>v</i> .
<i>N</i>	set of residue indexed by <i>n</i>	<i>W_{vessel_v}</i>	The mass of oil carried by vessel <i>v</i> .
<i>CP_c</i>	set of crude <i>c</i> allowed to product residue type	<i>VtankIN_{v,k,t}</i>	The feed mass from vessel <i>v</i> to tank <i>k</i> at period <i>t</i> .
<i>UJ_v</i>	set of crude types unloaded by tank <i>v</i>	<i>VtankOut_{k,m,t}</i>	The feed mass from tank <i>k</i> to CDU <i>m</i> at period <i>t</i> .
<i>BJ_v</i>	set of crude types in the bottom oil of tank <i>k</i> .	<i>Wtank_{k,t}</i>	The mass of tank <i>k</i> at period <i>t</i> .
Parameters		<i>VT_{j,m,t}</i>	The feed of crude <i>j</i> into CDU <i>m</i> at period <i>t</i> .
<i>C_{oilchange}</i>	The cost for a single crude oil blend formula change.	<i>P_{j,l}</i>	The property <i>l</i> of the crude <i>j</i> .
<i>C_{tankchange}</i>	The cost for a single tank change.	<i>R_{m,RES,t}</i>	The yield of residue for crude <i>j</i> in CDU <i>m</i> .
<i>C_{vessel}</i>	The waiting cost and unloading cost for the vessel <i>v</i> at sea.	<i>VCN_{m,n,t}</i>	The amount of residue <i>n</i> produced by CDU <i>m</i> at period <i>t</i> .
<i>FDmax_m</i>	the upper limit of daily feed into CDU <i>m</i> , which essentially equals the normal feed rate	<i>VC_{m,t}</i>	The total amount of residue produced by CDU <i>m</i> at period <i>t</i> .
<i>FDmin_m</i>	The lower limit of daily feed into CDU <i>m</i> , which allows for reducing the feed rate when normal feed cannot meet constraints.	<i>VCW_{m,t}</i>	The total amount of wax oil produced by CDU <i>m</i> at period <i>t</i> .
<i>P_{max,l,m}</i>	The upper limit of property <i>l</i> for the mixed feed into CDU <i>m</i> .	<i>VCNW_{m,n,t}</i>	The mass of wax oil produced when residue <i>n</i> is generated by CDU <i>m</i> at period <i>t</i> .
<i>P_{min,l,m}</i>	The lower limit of property <i>l</i> for the mixed feed into CDU <i>m</i> .	<i>R_{v,c,k}</i>	The similarity between crude <i>j</i> to be unloaded from vessel <i>v</i> and the bottom oil in tank <i>k</i> in terms of residue oil production type
<i>FT_{min,j,m}</i>	The lower limit daily feed of crude <i>j</i> to CDU <i>m</i>	<i>rand</i>	A random number generated from a uniform distribution in the range [0,1].
<i>FT_{max,j,m}</i>	The upper limit daily feed of crude <i>j</i> to CDU <i>m</i>	<i>X_{m,p,t}</i>	A random number for feeding tank <i>p</i> of CDU <i>m</i> in period <i>t</i> .
<i>FCNmax_{m,n}</i>	The upper limit daily residue <i>n</i> production by CDU <i>m</i>	<i>CF_{m,p,t}</i>	The oil discharge volume from feeding tank <i>p</i> of CDU <i>m</i> in period <i>t</i> .
<i>FCNmin_{m,n}</i>	The lower limit daily residue <i>n</i> production by CDU <i>m</i>		
<i>Limit_m</i>	The upper limit of crude types feed to device <i>m</i> simultaneously.		
<i>Wtankoutpunish_k</i>	the overload mass of entity tank <i>k</i> , with max value ≤ 1 % of tank mass upper limit.		
<i>PL_m^{min}</i>	The processing load lower limit of CDU <i>m</i>		

1.2. Previous research

Mathematical programming (MP) methods are widely applied in crude oil scheduling optimization. Typical mathematical programming models can be divided into two types: discrete-time and continuous-time models. In a discrete-time model, the scheduling period is divided into several time intervals, and binary variables are used to represent operational activities. This method facilitates handling complex scheduling problems, making it popular among many researchers (de Assis and Camponogara, 2016; Liao et al., 2018; Panda and Ramteke, 2019). However, when refining the time intervals, the number of binary variables increases significantly, leading to a rise in computational complexity (Saharidis et al., 2009). To manage the large number of binary variables and complex constraints, Jia and Ierapetritou decomposed the short-term scheduling problem of the entire refining process into three problems: crude oil operations, production unit scheduling,

and gasoline blending and distribution. They formulated these three problems as Mixed-Integer Linear Programming (MILP) models and solved them independently (Jia and Ierapetritou, 2004). Liao et al. established an MILP model based on a flow database for multi-product pipelines with multiple pump stations to optimize their detailed scheduling (Liao et al., 2018). Unlike discrete-time methods, continuous-time methods have uneven time intervals, allowing events to start and end at any point within the continuous time domain (Zhao et al., 2016). These models eliminate the constraints of predefined time intervals by incorporating time variables, significantly reducing the number of binary variables. However, the reduction of binary variables introduces nonlinear constraints into the model, resulting in a Mixed-Integer Nonlinear Programming (MINLP) model (Chen et al., 2012). Cafaro et al. proposed a MINLP continuous-time formulation to rigorously track the power consumption of each pipeline segment through nonlinear equations (Cafaro et al., 2015). Cerdá et al. proposed a

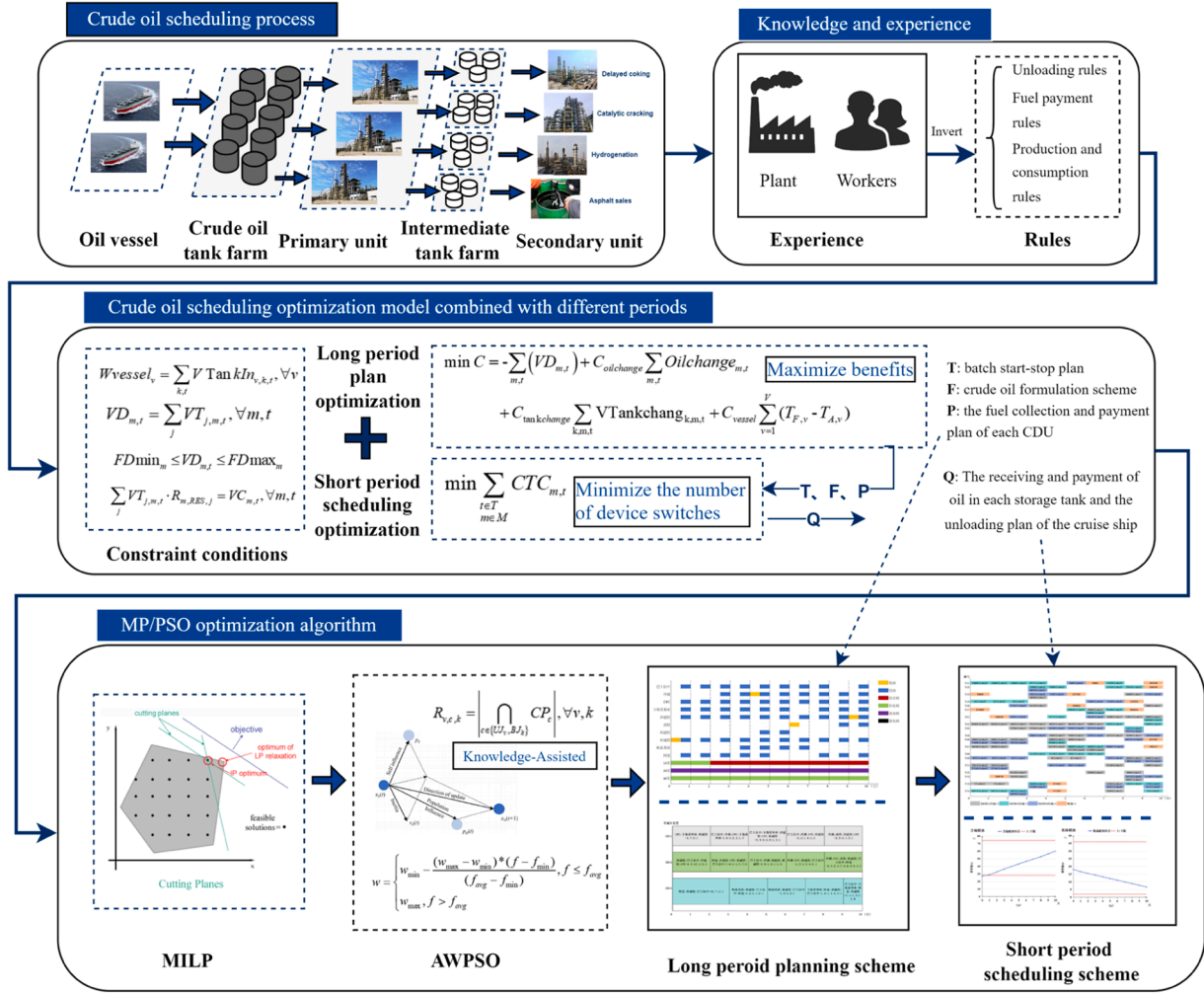


Fig. 1. Overall technical flowchart of the paper.

slot-based MINLP formulation that explicitly considers the True Boiling Point (TBP) curve of crude oil, thereby providing a series of suitable and stable feedstocks for the Crude Distillation Unit (CDU) (Cerdá et al., 2017). Additionally, when formulating scheduling problems using continuous-time models, it is necessary to know the number of operations to be executed in advance, thereby reducing its applicability. Moreover, it is nearly impossible to know the number of operations to be executed in advance when using continuous-time variables to formulate scheduling problems, which is practically challenging. To address the uncertainty of production yields in crude oil scheduling, Dai et al. employed the GIKSVC algorithm based on mathematical programming to construct uncertain sets, thereby transforming deterministic models into uncertainty models to effectively tackle the challenges posed by yield fluctuations (Dai et al., 2021). To address the uncertainty in oil properties during crude oil blending optimization, Dai et al. proposed a data-driven robust optimization method based on mathematical programming, which constructs uncertainty sets using principal component analysis and robust kernel density estimation, and integrates them into the optimization model to address fluctuations in crude oil properties during blending (Dai et al., 2020).

As (Li et al., 2007) states, existing commercial solvers may fail to find the optimal solution when solving MINLP problems. Therefore, the commonly used approach at present is to iteratively solve MILP and Nonlinear Programming (NLP) problems (Assis et al., 2021, 2019; Mouret et al., 2011). Hence, the application of mathematical programming techniques is challenging in practical scenarios (Floudas and Lin,

2004). It is worth mentioning that when formulating scheduling problems using mathematical programming models, people often overlook some practical constraints, resulting in schedules that are unrealistic or infeasible for actual refineries (Mendez et al., 2006).

In recent years, researchers have begun exploring heuristic and metaheuristic optimization algorithms to address this issue and apply them to scheduling problems (Branke et al., 2015; Gao et al., 2019; Li and Wang, 2022). Among these, the Particle Swarm Optimization (PSO) algorithm is a very popular metaheuristic algorithm for solving complex problems. Compared to other metaheuristic algorithms, PSO has fewer parameters and high computational efficiency, making it favored by many researchers (Jain et al., 2022; Meng et al., 2022; Periasamy et al., 2022; Zhu et al., 2011). For examples, H. Zhang et al. combines the MILP model with PSO to solve task scheduling problems in Digital Array Radar (DAR), fully utilizing capability of DAR to handle various tasks (Zhang et al., 2018). Ankita and Sahana proposed the Balanced Particle Swarm Optimization (Ba-PSO) to solve scheduling problems in computational grids (Ankita and Sahana, 2022). Lang and Zhao proposed an improved PSO algorithm for the well production scheduling problem of lightly loaded oil wells in oil field development. They also introduced a new velocity updating formula to approximate the solution to this problem (Lang and Zhao, 2016). To address the complexity of large-scale crude oil scheduling, Zhang et al. developed a heuristic rule-driven bi-level evolutionary algorithm based on meta-heuristic methods. By utilizing global search and local optimization, the algorithm improves solution feasibility and efficiency. Experimental results show that this method

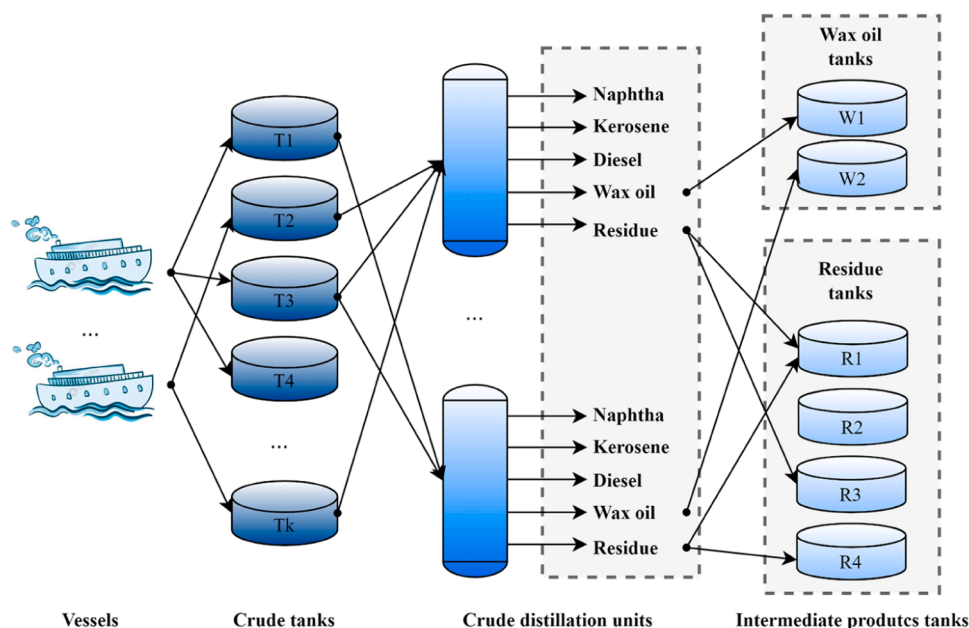


Fig. 2. Crude oil primary processing process.

effectively handles short-term scheduling problems in large-scale crude oil scheduling (Zhang et al., 2024).

Despite significant progress in crude oil scheduling optimization achieved by existing research, there are still some shortcomings. Firstly, traditional rule-based models demonstrate certain limitations in dealing with complex and dynamically changing real-world situations, making it difficult to achieve true optimization. Secondly, standalone mathematical programming methods face challenges of long solution times and high computational resource consumption when solving large-scale scheduling problems, while standalone metaheuristic algorithms, although capable of improving solution speed, still require improvements in solution accuracy and stability.

Building upon existing research, this paper proposes a knowledge-integrated planning and scheduling combined crude oil scheduling model, achieving integrated optimization solutions through a mixed optimization algorithm (MP/PSO). This method not only effectively addresses large-scale crude oil scheduling problems but also enhances solution speed and optimization effectiveness while ensuring solution accuracy.

1.3. Motivation and contributions

Based on the above discussions, this paper aims to study how to integrate long-term and short-term crude oil scheduling. After optimizing a reasonable long-term plan, further optimization is done to decompose it into short-term scheduling plans. We propose a knowledge-experience-based long-term and short-term combined scheduling optimization method to obtain feasible scheduling for practical cases. The contributions of this paper are as follows:

- (1) Propose a knowledge-experience-based optimization strategy combining long-term and short-term periods, achieving operational short-term scheduling optimization based on resource-based long-term planning results.
- (2) Introduce a hybrid optimization algorithm combining mathematical programming and particle swarm optimization to achieve feasible scheduling within a reasonable time.
- (3) Successfully validate the proposed method with real-case applications.

The remainder of this paper is organized as follows: [Section 2](#) provides the background information on crude oil scheduling problems; [Section 3](#) introduces the model and solution algorithm for crude oil scheduling; [Section 4](#) reports experimental results and performance analysis of real cases; finally, [Section 5](#) concludes the paper. The overall technical roadmap of this paper is illustrated in [Fig. 1](#).

2. Problem description

2.1. Crude oil primary processing process

The journey from crude oil to the final refined petroleum products encompasses three distinct stages: primary processing, secondary processing, and storage/transportation of the refined products. In the primary processing phase, crude oil undergoes fractionation yielding a spectrum of distillates including light oil, straight-run gasoline, kerosene, light diesel or heavy diesel fractions, alongside diverse lubricating oil fractions (Yang et al., 2023). These components are meticulously refined and blended to conform to industry standards, rendering them market-ready.

The primary processing of crude oil involves three phases, as illustrated in [Fig. 2](#):

- (1) **Crude oil unloading phase:** Vessels or pipelines transport crude oil to designated storage areas for temporary containment.
- (2) **Transportation phase:** Crude oil is conveyed to refineries via pipelines or other designated transportation modalities. Refineries exercise discernment in selecting one or more varieties of crude oil for amalgamation, a strategic endeavor aimed at fulfilling requisite chemical specifications and bespoke performance thresholds.
- (3) **Supply phase:** Upon ingress into the refinery, crude oil is subjected to a process of fractionation, yielding an array of chemical constituents, encompassing light oil, heavy oil, kerosene, lubricating oil, and sundry other petroleum derivatives. This supply process entails a multifaceted sequence of operations, spanning distillation, catalysis, cracking, and an array of filtration, separation, and pressurization procedures.

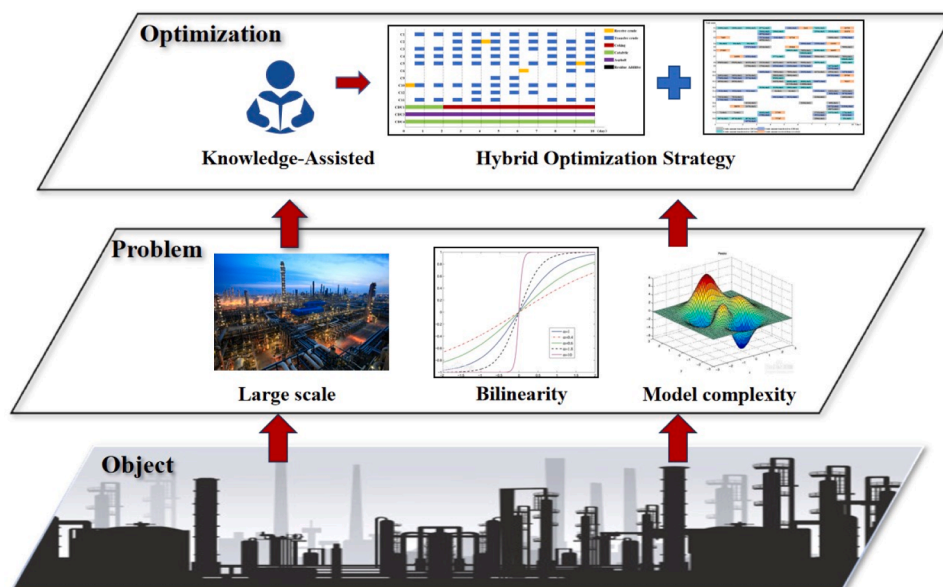


Fig. 3. Diagram illustrating scheduling issues in coastal refinery enterprises.

2.2. Crude oil scheduling problem

In petrochemical enterprises, production scheduling plans typically comprise long-term scheduling (monthly or every ten days), short-term planning (weekly, daily, or hourly), and real-time scheduling. The long-term production scheduling plan of a refinery aims to maximize the enterprise's profit. It is based on market demand and prices for various products, characteristics of existing crude oil, and the status of each production unit in the refinery. It determines the types of crude oil to be processed over a longer period, the quantity to be processed, the blending of raw materials and intermediate products in each production process, and rough crude oil procurement plans. Short-term production scheduling plans deal with dynamic systems. Unlike long-term plans, the details of the production process such as when to process which raw materials, at what rate, where to transport them, and how much to transport are determined by short-term production plans, which are not considered in long-term production plans (Wu et al., 2005).

Taking a typical coastal refinery enterprise as an example, the crude oil scheduling problem is depicted in Fig. 3. Below, we elaborate on the initial conditions, constraints, objectives, decisions, and characteristics of the problem.

2.2.1. Object layer

As a large-scale industrial production system, the scheduling optimization objects of an oil refinery mainly include processes such as crude oil procurement, transportation, storage, and processing. The gray area at the bottom illustrates the outline of a refinery, specifying the specific objects and environment for optimization and scheduling, thereby providing initial conditions and constraints.

Initial conditions: This includes, among other things, the time of arrival of the tanker at the port, the amount of crude oil loaded and the rate at which it is being discharged, as well as the type and volume of crude oil in the tanks and pipelines, and the current status of the equipment in the system.

Constraints: These include resource constraints, such as limitations on the number and capacity of tanks, limitations on the rate of pipeline transmission, and the type and level of initial crude oil inventories; and process constraints, including the inability of tanks to conduct simultaneous loading and unloading operations, the need for certain crude oils to be stored and transported in compliance with specific safety regulations, and the need for crude distillation units to be able to

consistently process specific types of crude oils.

2.2.2. Problem layer

Listed them as issues to be overcome in the scheduling process:

Large scale: The large size of the system and the many variables and constraints involved lead to a complex scheduling problem.

Bilinearity: The variable relationships in scheduling are complex, and nonlinear and bilinear relationships increase the complexity and difficulty of solving the problem.

Model complexity: The high complexity of scheduling models, which may involve multiple dimensions and complex mathematical expressions, makes problem solving more difficult.

2.2.3. Optimization layer

To cope with the above problems, this paper proposes a knowledge-assisted integrated planning and scheduling crude oil scheduling optimization model that combines long-term planning optimization and short-term scheduling optimization:

Knowledge-assisted: The global search phase of the optimization process is accelerated by using the experience and knowledge of domain experts to generate well-performing initial populations through heuristic rules.

Hybrid optimization strategy: Combining mathematical programming and particle swarm optimization to solve large-scale, bilinear, and complex models through a combination of global search and local optimization.

Long-term planning optimization: The goal is to minimize operating and transportation costs and maximize refining margins while meeting constraints such as material balance, crude oil properties, sideline throughput, recipe switching, and wax-residue inventory limits.

Short-term scheduling optimization: Using the initial formulation obtained from the long-term optimization as an input condition, the optimization objective is to minimize the number of unit switching and improve the operability and efficiency of scheduling (Chunpeng and Gang, 2009).

In this paper, we effectively cope with the complexity and dynamics faced by modern refineries in the scheduling process by constructing a knowledge-assisted optimization model for integrated planning and scheduling of crude oil scheduling. By combining the optimization strategies of long-term planning and short-term scheduling, and using knowledge-assisted and hybrid optimization algorithms, the efficiency

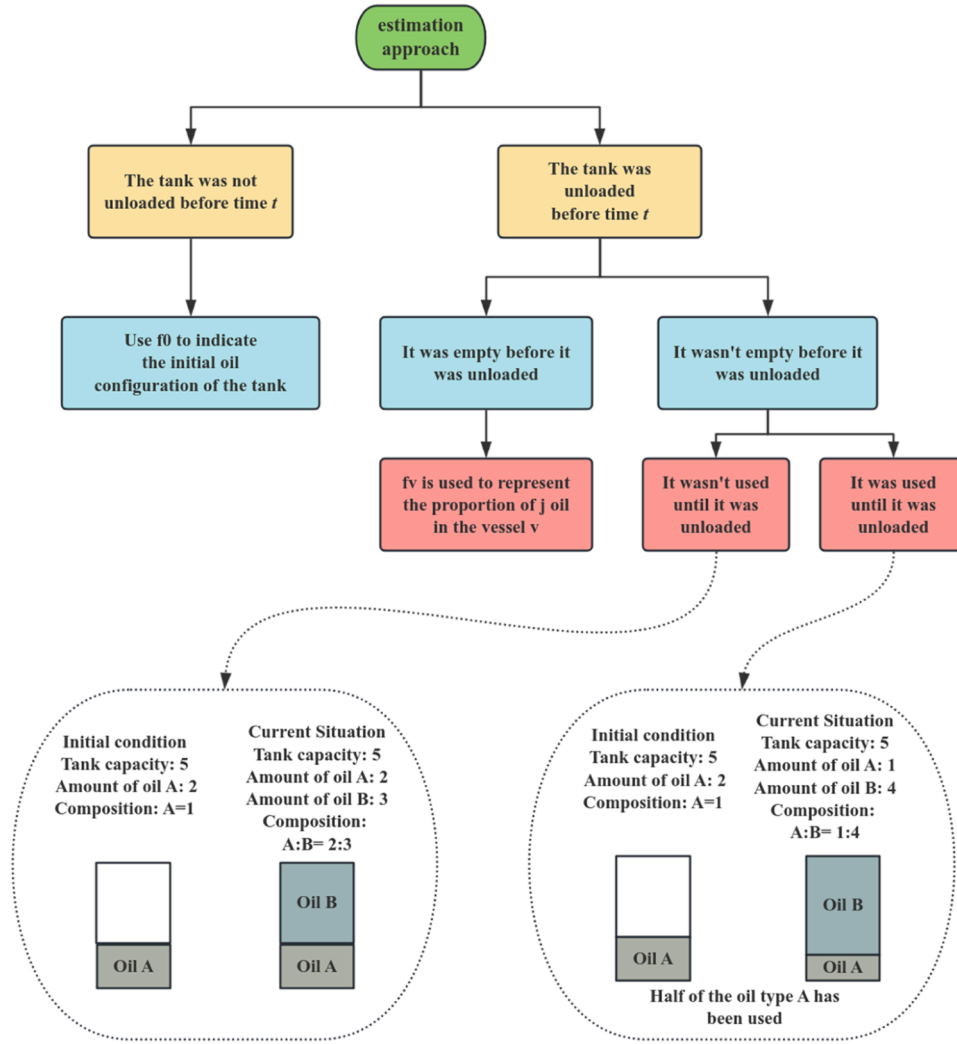


Fig. 4. Estimation idea specification.

and effectiveness of scheduling are significantly improved.

3. Model establishment and solution

For the readability of the article, detailed descriptions of the symbols in the following formulas are listed in the **Notation** section at the end of the document.

3.1. Crude oil long-term scheduling model

3.1.1. Optimization objective

The objective function for the long-term crude oil planning model can be expressed as Eq. (1). Among them, m represents the index of CDU, with the m -th CDU corresponding to the set M ; t is the time period index, with a time scale in days corresponding to the set T . $VD_{m,t}$ represents the oil intake rate of the atmospheric and vacuum distillation unit, indicating that the production load of the primary unit should be as high as possible to generate higher revenue. $C_{oilchange}$ represents the cost of switching oil blend formulas once, while $Oilchange$ indicates when the oil blend formula changes. $C_{tankchange}$ is the cost of switching tanks once, and $VTankchange$ indicates when the tank switch occurs. These two optimization objectives aim to minimize the operational costs resulting from oil blend formula and tank switching. C_{vessel} represents the waiting and unloading costs of vessel v at sea, $T_{A,v}$ represents the vessel's oil arrival time, and $T_{F,v}$ represents the unloading completion time. The goal

is to minimize the vessel's waiting and unloading costs.

$$\min C = -\sum_{m,t} (VD_{m,t}) + C_{oilchange} \sum_{m,t} Oilchange_{m,t} + C_{tankchange} \sum_{k,m,t} VTankchange_{k,m,t} + C_{vessel} \sum_{v=1}^V (T_{F,v} - T_{A,v}) \quad (1)$$

3.1.2. Conventional constraint

(1) Material Balance for Ships:

$$W_{vessel,v} = \sum_{k,t} VTankIn_{v,k,t}, \forall v \quad (2)$$

$$10000 \cdot XTankIn_{v,k,t} \leq VTankIn_{v,k,t} \leq 1000000 \cdot XTankIn_{v,k,t}, \forall v, k, t \quad (3)$$

Eq. (2) indicates that the total amount of oil carried by all ships is equal to the sum of the weights each ship transfers to the tanks. Eq. (3) states that in order to reduce the number of switches for unloading from physical tanks, at least ten thousand tons should be unloaded from physical tanks.

(2) Material Balance for Physical Tanks:

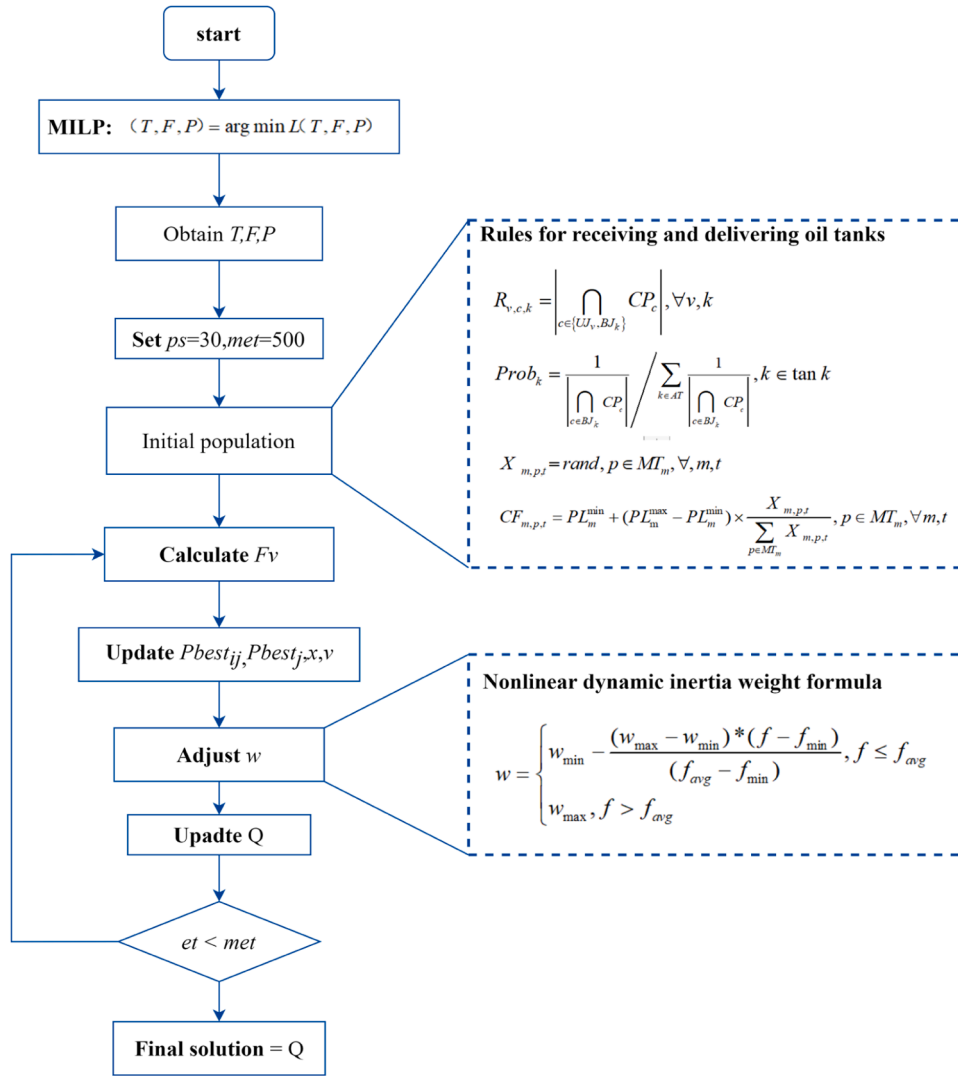


Fig. 5. MP/PSO algorithm flowchart.

$$WTank_{k,t} = WTank0_k + \sum_{v,t'} VTankIn_{v,k,t'} - \sum_{m,t'} VTankOut_{k,m,t'}, \quad \forall k, t, t' < t \quad (4)$$

$$WTankOutPunish_k \leq 0.01 \cdot WTankMax_k, \forall k \quad (5)$$

$$0 \leq WTank_{k,t} \leq WTankMax_k + WTankOutPunish_k, \forall k, t \quad (6)$$

In Eq. (4), $WTank_{k,t}$ represents the mass of tank k at time t , which equals the initial mass of tank k plus the difference between the inflow and outflow of the tank up to time t . In Eq. (5) and (6), the slack variable $WTankOutPunish_k$ represents the overload mass of tank k , with its maximum value not exceeding 1 % of the tank's upper mass limit.

(3) Feed Balance for Atmospheric Distillation Units:

$$VD_{m,t} = \sum_j VT_{j,m,t}, \forall m, t \quad (7)$$

$$FD_{\min} \leq VD_{m,t} \leq FD_{\max}_m \quad (8)$$

Eq. (7) indicates that the feed amount of CDU m at time t is equal to the sum of the feed amounts of each type of oil input into

CDU m at time t . In Eq. (8), $FD_{\max}_m$ represents the upper limit of the daily feed for unit m , which is essentially the normal feed amount. $FD_{\min}_m$ represents the lower limit of the daily feed for unit m , which allows the feed amount to be reduced if the normal feed cannot meet the constraint.

(4) Feed Properties Constraints:

$$P_{\min,l,m} \cdot VD_{m,t} \leq \sum_j VT_{j,m,t} \cdot P_{j,l} \leq P_{\max,l} \cdot VD_{m,t}, \forall m, l, t \quad (9)$$

Eq. (9) represents the upper and lower limits of the feed properties of CDU m .

(5) Rate Limits from Virtual Tanks to Atmospheric Distillation Units:

$$FT_{\min,j,m} \cdot XT_{j,m,t} \leq VT_{j,m,t} \leq FT_{\max,j,m} \cdot XT_{j,m,t}, \forall j, m, t \quad (10)$$

(6) Residue Oil Yield Constraints:

$$\sum_j VT_{j,m,t} \cdot R_{m,RES,j} = VC_{m,t}, \forall m, t \quad (11)$$

$$\sum_n VCN_{m,n,t} = VC_{m,t}, \forall m, t \quad (12)$$

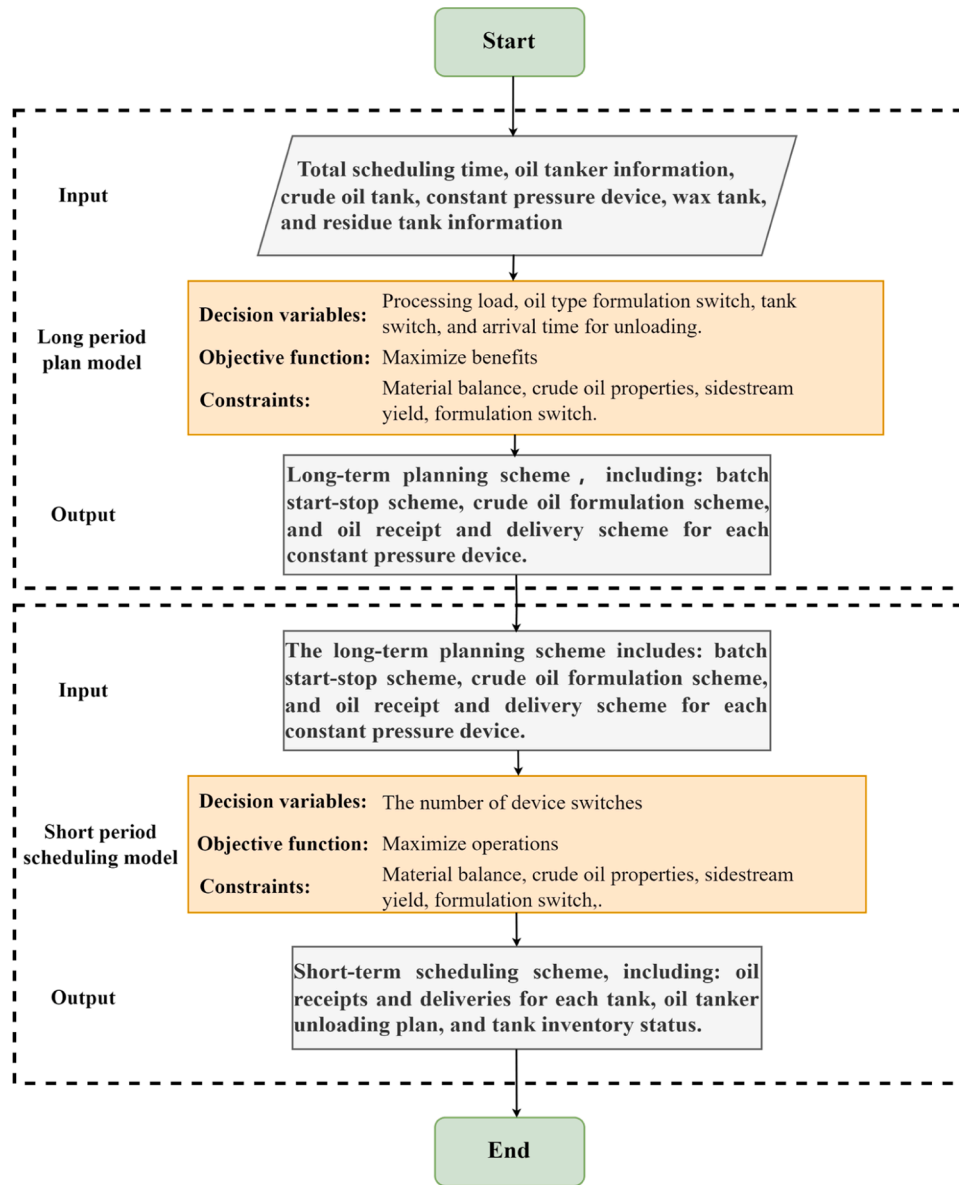


Fig. 6. Solution Process Diagram Illustrating Information Flow Between Long-Term and Short-Term Scheduling.

Table 1
Crude oil properties information.

Crude	Sulfur Content(%)	TAN(mgKOH/g)	Density(g/cm3)	Types of Residual Oil Produced			
				Coking	Catalytic	Asphalt	Residue Additive
C1	3.32	0.17	0.886	1	0	1	1
C2	3.42	1.52	0.916	1	0	1	1
C3	0.55	0.09	0.804	1	0	1	1
C4	2.00	0.48	0.939	1	0	1	0
C5	1.00	0.05	0.827	1	0	1	1
C6	1.85	0.06	0.86	1	0	1	1
C7	0.29	0.08	0.865	1	1	1	0
C8	0.30	0.55	0.891	0	1	0	0
C9	0.17	0.05	0.813	1	0	1	1
C10	2.78	0.18	0.875	1	0	1	1
C11	0.38	1.48	0.907	1	0	0	1
C12	1.43	0.6	0.873	1	0	0	1

$$VCN_{m,n,t} \leq XC_{m,n,t} \cdot FCN_{\max,m,n} \quad (13)$$

$$VCN_{m,n,t} \geq 100 \cdot XC_{m,n,t} \quad (14)$$

In Eq. (11), $R_{m,RES,j}$ represents the yield of residual oil from the j oil in

Table 2
Crude oil yield information.

CDU	Products(wt%)	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
CDU1	Naphtha	19.5	17.6	30	13.4	28.9	19.2	15.81	10.45	35.5	17.6	9.6	16.75
	Kerosene	7.1	3.7	20.9	3.9	9.4	9.2	6.5	6.4	13.7	7.5	8.7	6.8
	Diesel	17.28	15.54	24.72	16.22	22.89	15.35	17.88	14.86	26.16	18.43	19.7	17.88
	Wax Oil	26.71	24.11	18.51	28.31	22.81	29.41	31.88	32.08	19.61	24.61	30.7	26.97
	Scrubbing	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
	Residue	27.21	34.91	2.11	35.31	12.71	23.61	23.74	32.24	1.91	28.81	29.7	26.25
CDU3	Naphtha	17	16.1	30	12.34	26.82	17.54	15.81	10.45	35.5	16.16	9.6	16.75
	Kerosene	8.7	3.9	20.9	4.1	10	9.6	6.5	6.4	13.7	8	8.7	6.8
	Diesel	16	15.54	24.72	16.32	22.79	15.37	17.88	14.86	26.16	18.43	19.7	17.88
	Wax Oil	25.87	23.57	18.51	27.98	22.27	28.87	31.88	32.08	19.61	24.07	30.7	26.97
	Scrubbing	3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7
	Residue	26.8	33.65	2.11	34.54	11.45	22.35	23.74	32.24	1.91	27.55	29.7	26.25
CDU4	Naphtha	17	16.32	30	12.34	26.82	17.65	15.81	10.45	35.5	16.25	9.6	16.77
	Kerosene	8.7	3.9	20.9	4.1	10	8	6.5	6.4	13.7	8	8.7	6.8
	Diesel	16	15.54	24.72	16.32	22.79	16.97	17.88	14.86	26.16	18.43	19.7	17.85
	Wax Oil	26.71	23.78	18.51	27.98	22.27	29.08	31.88	32.08	19.61	24.28	30.7	27.18
	Scrubbing	3	3	3	3	3	3	3	3	3	3	3	3
	Residue	26.8	34.14	2.11	34.54	11.45	22.84	23.74	32.24	1.91	28.04	29.7	26.74

Table 3
Wax and residue oil tank original information.

Tank Name	Initial Inventory (t)	Capacity(t)[Lower, Upper]	Consumption(t)[Lower, Upper]	Fixed Output (t)	Fixed Consumption (t)
W1	27,184	[16,000,74,000]	0	5400	17,286
W2	18,076	[2000,28,000]	0	0	1107
R1	17,780	[13,000,34,000]	[8880,11,040]	720	480
R2	28,419	[8000,40,000]	[1920,1920]	0	0
R3	19,335	[10,000,45,000]	[3120,3120]	0	0
R4	7699	[1000,8000]	[5520,5520]	480	0

Table 4
CDU processing data information.

Category	Type name	CDU1 Lower/ Upper	CDU3 Lower/ Upper	CDU4 Lower/ Upper
Properties	Sulfur Content (%)	1/2.7	2/3	0/2.7
	TAN (mgKOH/g)	0/0.5	0/0.5	0/2
	Density (g/cm3)	0.84/0.885	0.84/0.880	0.84/0.891
Yield	Naphtha (t/day)	1440/5280	1440/4080	2640/7200
	Kerosene (t/day)	1200/3360	1680/3120	960/3120
	Diesel (t/day)	2400/5280	2880/5280	3840/9600
	Wax Oil (t/day)	3120/5760	3600/6240	4800/ 10,800
	Scrubbing (t/day)	0/1000	0/1000	0/1000
	Residue (t/day)	4080/6480	4320/6000	4800/7440
	Processing Load (t/ day)	18,500/ 18,500	20,500/ 20,500	20,000/ 25,000

Table 5
The refinery scheduling parameters of Case1 and Case2.

	Horizon	Parcels	Crude oil storage tanks	CDUs	Crude types	Residue types
Case1	4 days	1	6	3	12	4
Case2	10 days	4	20	3	12	4

Table 6
Problem Sizes of Case 1 and Case 2 and Computational Resources Required for Different Methods.

	MP				PSO		MP/PSO	Manual Scheduling
	Total var.	Binary var.	Total cons.	CPU time(s)	Decision var.	CPU time(s)	CPU time(s)	Scheduling Time (days)
Case1	1601	758	4990	5.36	240	24.48	29.84	3–7
Case2	20,961	8902	63,786	200.02	600	103.86	303.88	3–7

CDU m . In Eq. (12), $VCN_{m,n,t}$ represents the amount of residual oil n produced by the atmospheric and vacuum distillation CDU m at time t , and $VC_{m,t}$ represents the total residual oil produced by CDU m at time t .

Eqs. (13) and (14) represent the upper and lower limits of the residual oil n produced by CDU m at time t , respectively. $XC_{m,n,t}$ is a 0–1 variable.

Table 7
Tank information (Case 1).

Tanker Name	Oil Type	Cargo Capacity (t)	Arrival Time (days)
Tanker 1	C10	100,000	1

Table 8
Original tank information (Case1).

Tank Name	Initial Inventory(Oil Type: Oil Volume (t))	Liquid Level Limit (m) [Lower, Upper]	Tank Bottom Area(m ²)
T9	[C2:37,884]	[2.4,19.3]	5024
T10	[C3:17,443.3]	[2.4,19.3]	5024
T12	[C5:52,329.7] [C1:24,736.8] [C10:50,223.2]	[2.4,19.3]	5024
T13	[C11:39,800]	[2.6,19.3]	5024
T18	[C1:21,949]	[2.4,16.4]	2826
T19	[C12:80,272]	[2.4,19.3]	5024

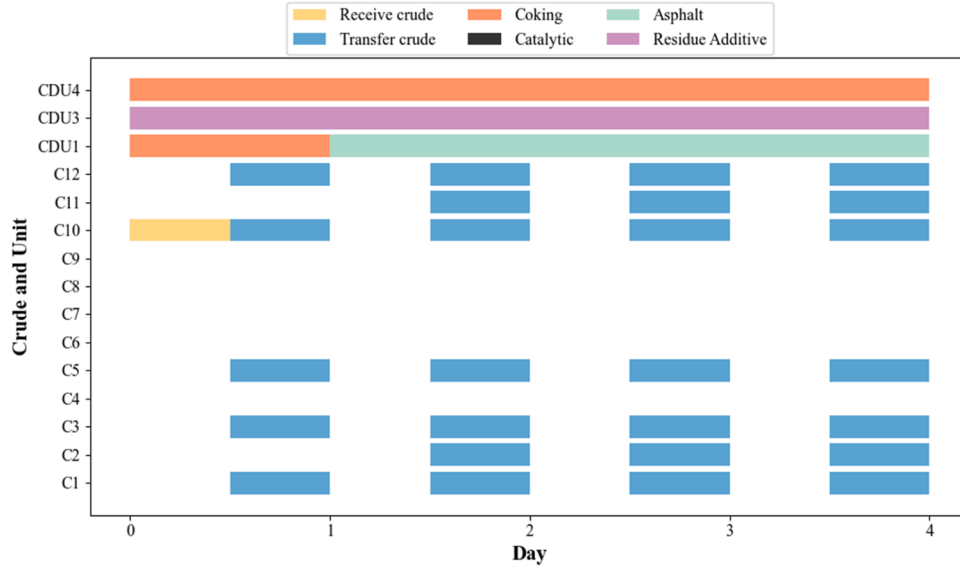


Fig. 7. Long period planning scheme Gantt chart (Case 1). C1-C12 represent the virtual tanks for the oil types, while CDU1, CDU3, and CDU4 represent the residue oil output plans for each respective unit.

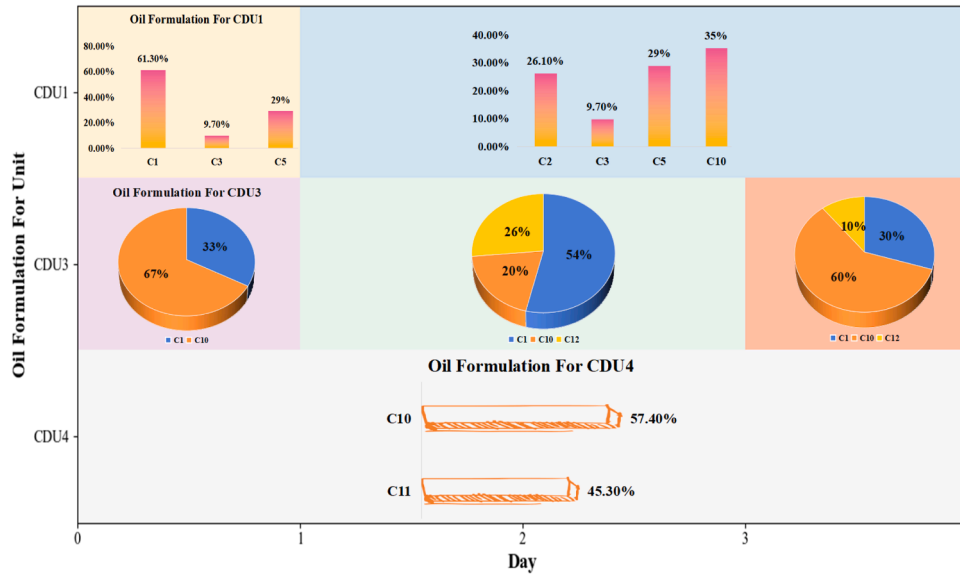


Fig. 8. Long period processing formula (Case 1). Bar chart, pie chart, and column chart respectively represent the oil blend formulations for CDU1, CDU3, and CDU4.

(1) Wax Oil Yield Constraints:

$$\sum_j VT_{j,m,t} \cdot R_{m,WAX,j} = VCNW_{m,t}, \forall m, t \quad (15)$$

$$\sum_n VCNW_{m,n,t} = VCNW_{m,t}, \forall m, t \quad (16)$$

$$XC_{m,n,t} \cdot FCNmin_{m,n} \leq VCNW_{m,n,t} \leq XC_{m,n,t} \cdot FCNmax_{m,n}, \forall m, n, t \quad (17)$$

Eqs. (15-17) are similar to Eqs. (11-14) and will not be elaborated on here.

(2) Recipe change calculation:

$$XT_{j,m,t-1} - XT_{j,m,t} \leq OilChange_{m,t}, \forall m, t \quad (18)$$

Eq. (18) is used to determine whether a recipe change occurs; if it does, $OilChange_{m,t}$ is set to 1.

3.1.3. Operating constraint

(1) Unloading Rules: Defines the rules and constraints for unloading crude oil from tankers into storage tanks

$$\sum_k Arrive_v \cdot XTankIn_{v,k,t} = \sum_k t \cdot XTankIn_{v,k,t}, \forall v, t \quad (19)$$

$$\sum_{k,t} XTankIn_{v,k,t} \leq 2, \forall v \quad (20)$$

$$\sum_{v,t} XTankIn_{v,k,t} \leq 1, \forall k \quad (21)$$

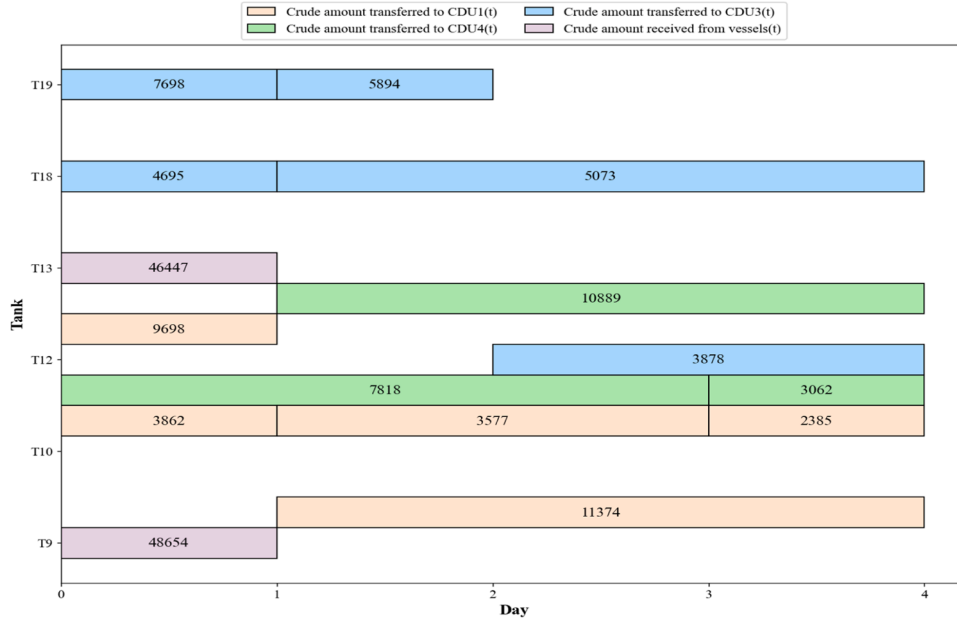


Fig. 9. Short-period scheduling scheme Gantt chart (Case 1).

Table 9

Tank information (Case 2).

Tanker Name	Oil Type	Cargo Capacity (t)	Arrival Time (days)
Tanker 1	C10	100,000	1
Tanker 2	C2	140,000	5
Tanker 3	C6	94,000	7
Tanker 4	C5	125,000	10

Table 10

Original tank information (Case2).

Tank Name	Initial Inventory(Oil Type: Oil Volume (t))	Liquid Level Limit (m) [Lower, Upper]	Tank Bottom Area(m ²)
T1	[C1:7577.3]	[2.4,17.4]	1512
T2	[C10:15,154.7]	[2.4,17.4]	1512
T3	[C12:21,994]	[2.4,13.5]	1292
T4	[C10:7593]	[2.4,13.5]	1292
T5	[C1:5918]	[2.4,17.4]	2826
T6	[C5:31,963]	[2.4,17.4]	2826
T7	[C10:3746]	[2.4,19.3]	5024
T8	[C12:1873]	[2.4,19.3]	5024
T9	[C1:9992.5]	[2.4,19.3]	5024
T10	[C10:45,521.5]	[2.4,19.3]	5024
T11	[C4:60,512]	[2.4,19.3]	5024
T12	[C2:37,884]	[2.4,19.3]	5024
T13	[C3:17,443.3]	[2.4,19.3]	5024
T14	[C5:52,329.7]	[2.4,19.3]	5024
T15	[C1:65,703.3]	[2.4,19.3]	5024
T16	[C10:18,531.7]	[2.6,19.3]	5024
T17	[C1:24,736.8]	[2.6,19.3]	5024
T18	[C10:50,223.2]	[2.6,19.3]	5024
T19	[C11:39,800]	[2.6,19.3]	5024
T20	[C3:62,733]	[2.6,19.3]	5024
T21	[C9:35,050]	[2.4,16.4]	2826
T22	[C3:11,744]	[2.4,17.4]	2826
T23	[C5:11,744]	[2.4,17.4]	2826
T24	[C4:5318]	[2.4,17.4]	2826
T25	[C1:21,949]	[2.4,16.4]	2826
T26	[C12:80,272]	[2.4,19.3]	5024
T27	[C7:3687.5]	[2.4,19.3]	5024
T28	[C8:3687.5]	[2.4,19.3]	5024

$$\sum_v X_{\text{TankIn}}_{v,k,t} + \sum_m X_{\text{TankOut}}_{k,m,t} \leq 1 \quad (22)$$

In Eq. (19), $X_{\text{TankIn}}_{v,k,t}$ indicates whether ship v unloads oil to tank k at time t , aiming to reduce the tanker's waiting costs by unloading on the same day it arrives. Eqs. (20) and 21 state that a ship can only unload to two tanks, and each tank can only receive oil from one ship. This rule helps avoid multiple changes in the composition within a tank; if a tank is unloaded multiple times, it indicates that the utilization of other tanks is relatively low. Eq. (22) states that a tank cannot simultaneously receive and discharge oil.

(2) Storage Tank Oil Delivery Rules:

$$1920 \cdot X_{\text{TankOut}}_{k,m,t} \leq V_{\text{TankOut}}_{k,m,t} \leq 100000 \cdot X_{\text{TankOut}}_{k,m,t}, \forall k, m, t \quad (23)$$

a) Eq. (23) limits the oil delivery rate from storage tanks to the CDU.

$$\sum_k X_{\text{TankOut}}_{k,m,t} \leq 5, \forall m, t \quad (24)$$

b) Eq. (24) specifies that a single CDU can receive oil from up to five tanks simultaneously.

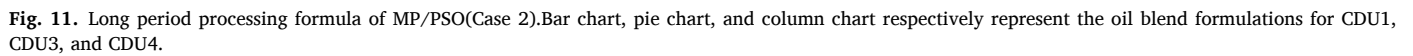
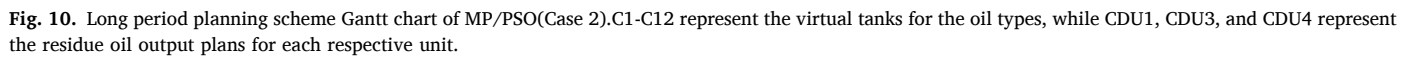
$$\sum_m X_{\text{TankOut}}_{k,m,t} \leq 2, \forall k, t \quad (25)$$

c) Eq. (25) specifies that a single tank can deliver oil to a maximum of two CDUs simultaneously.

(3) Oil Type Quantity Restriction Rules:

$$1 \leq \sum_j X_{T_j,m,t} \leq \text{Limit}_m, \forall m, t \quad (26)$$

There is no restriction on the number of virtual tanks supplying the units. However, according to the long-term schedule, there may be in-



In Eq. (27), $VT_{k,m,t}$ represents the oil transfer amount (mass) from

2) When a tank is discharged, it should be filled as much as possible. This rule is designed to help estimate the composition of the tank and

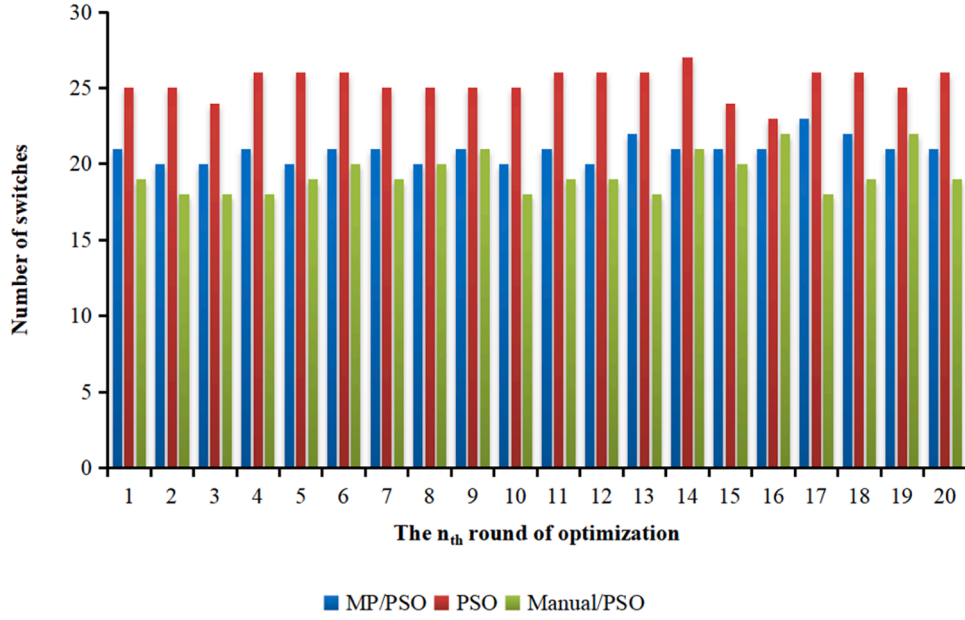


Fig. 12. MP/PSO, PSO and Manual/PSO switching times within 20 rounds.

Table 11

Number of constraint violations for MP/PSO, PSO and Manual/PSO within 20 rounds.

Optimization of the n_{th} round	MP/PSO	Manual/PSO	PSO	Optimization of the n_{th} round	MP/PSO	Manual/PSO	PSO
1	0	2	55	11	0	2	48
2	1	2	48	12	0	0	55
3	0	3	55	13	2	0	45
4	0	0	42	14	0	2	53
5	0	18	51	15	0	5	52
6	0	2	51	16	0	2	51
7	0	0	58	17	0	2	43
8	0	2	44	18	0	2	48
9	0	6	50	19	0	2	54
10	0	2	44	20	0	2	48

Table 12

Indicators obtained after 20 rounds of optimization using MP/PSO compared to PSO and Manual/PSO.

Indicators	MP/PSO	PSO	Manual/PSO
mean±std	20.9±0.768	25.35±0.910	19.3±1.23
FR	90 %	0 %	20 %

ensure that the number of times the discharge location is switched is minimized.

Estimation approach:

- 1) If the tank has not been discharged before time t , use $f0_{k,j}$ to represent the initial oil configuration of the tank, indicating the proportion of oil type j in tank k .
- 2) If the tank has been discharged before time t and was empty before discharging, use $f_{v,j}$ to represent the proportion of oil type j carried by the vessel v .
- 3) If the tank has been discharged before time t and was not empty before discharging, and has not been used prior to discharging.
- 4) If the tank has been discharged before time t and was not empty before discharging, and has been used prior to discharging.

Take approach 2) as an example for model implementation.

$$XInxEmpty_{v,k,t} = XTankIn_{v,k,t} \cdot XInxEmpty_{k,Arrive_v} \quad (28)$$

$$VTankOutBias_{v,k,m,t} = VTankOut_{k,m,t} \cdot XInxEmpty_{v,k,t} \quad (29)$$

$$VT_{k,m,t} = \sum_k VTankOut_{k,m,t} \cdot f0_{k,j} + \sum_k \sum_v VTankOutBias_{v,k,m,t} \cdot (f_{v,j} - f0_{k,j}) \quad (30)$$

In Eq. (28), the multiplication of two 0–1 binary variables is used to determine: $XTankIn_{v,k,t}$ determines whether entity tank k is being unloaded by vessel v at time t , and $XInxEmpty_{k,Arrive_v}$ whether entity tank k is empty when being unloaded by vessel v . In Eq. (29), $VTankOut_{k,m,t}$ represents the output quality from entity tank k to device m at time t . An additional constraint is introduced in the model to linearize the bilinear term by adding a bias variable:

$$VT_{k,m,t} = v, j \sum_k VTankOut_{k,m,t} \cdot f0_{k,j} + \sum_{v,k} VTankOutBias_{v,k,m,t} \cdot (f_{v,j} - f0_{k,j}) + \sum_{v,k} VTankOutBias_{2,v,k,m,t} \cdot (fd1_{v,k,j} + fd2_{v,k,j} + fd3_{v,k,j} + fd4_{v,k,j}) + \sum_{v,k} VTankOutBias_{3,v,k,m,t} \cdot (fd1_{v,k,j} + fd2_{v,k,j}), \forall j, m, t \quad (31)$$

In Eq. (31), $VT_{k,m,t}$ represents the oil (mass) delivered from tank k to the atmospheric distillation unit m at time t ; $f_{v,j}$ represents the proportion of oil j carried by vessel v ; $f0_{k,j}$ represents the initial proportion of oil j in entity tank k ; $fd1_{v,k,j}$ represents the reduction in the proportion of oil j when vessel v unloads into tank k (assuming the tank is unused and fully unloaded); $fd2_{v,k,j}$ represents the increase in the proportion of oil j when

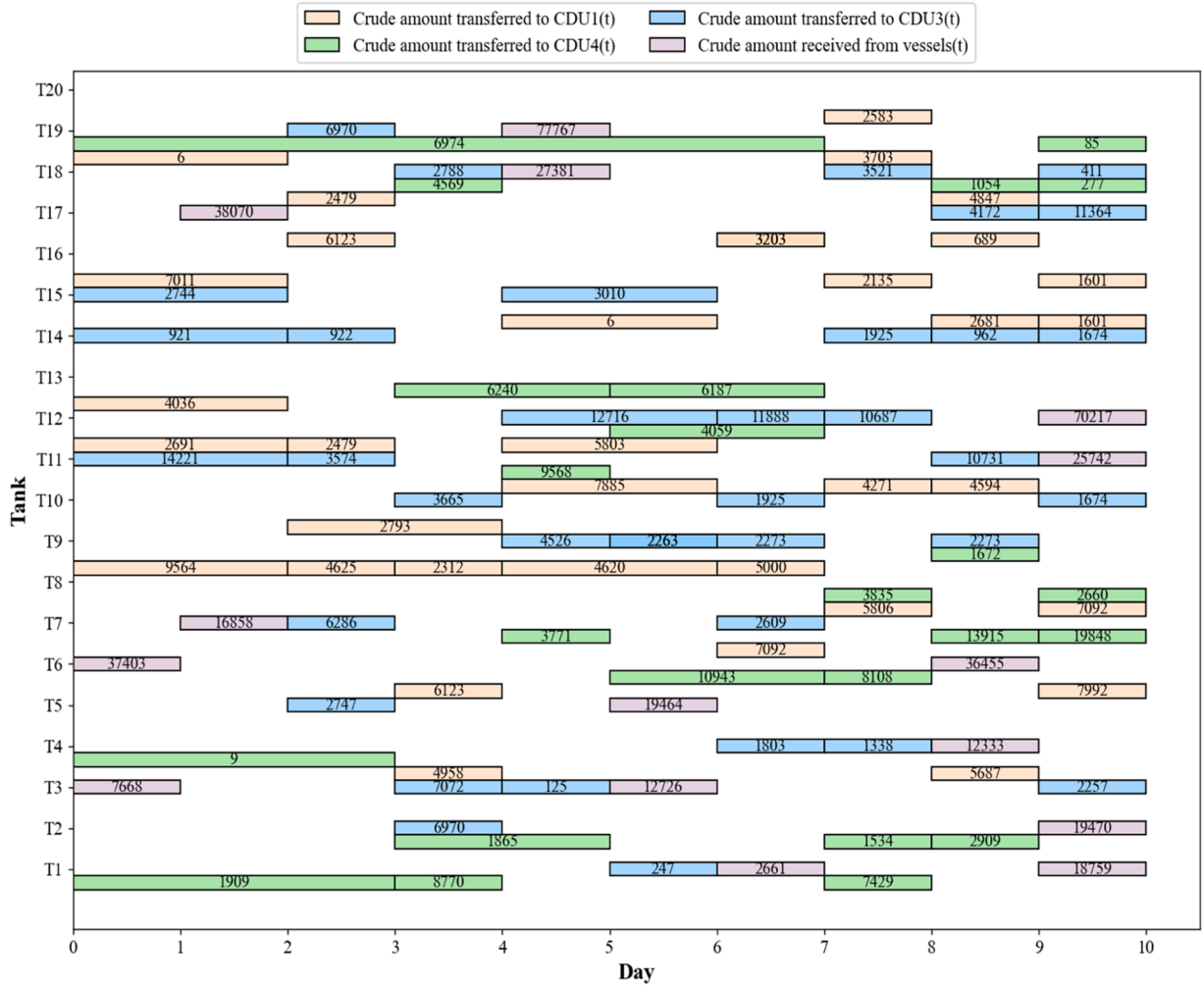


Fig. 13. Short-period scheduling scheme Gantt chart (MP/PSO).

vessel v unloads into tank k (assuming the tank is unused and fully unloaded); $fd3_{v,k,j}$ represents the reduction in the proportion of oil j when vessel v unloads into tank k based on $fd1_{v,k,j}$ and $fd2_{v,k,j}$ (assuming the tank has been used but is not empty and is fully unloaded); $fd4_{v,k,j}$ represents the increase in the proportion of oil j when vessel v unloads into tank k based on $fd1_{v,k,j}$ and $fd2_{v,k,j}$ (assuming the tank has been used but is not empty and is fully unloaded).

3.2. Crude oil short-term scheduling model

3.2.1. Optimization objective

Considering the short-term scheduling optimization method for operations, its objective is to ensure that the crude oil entering the atmospheric distillation units meets the quality standards and that the entire processing operation runs continuously and smoothly while satisfying the constraints of the long-term storage and transportation scheduling process. Additionally, it aims to minimize the number of switches between physical tanks and production processing units. The objective function is expressed as Eq. (32):

$$\min \sum_{\substack{I \in T \\ m \in M}} CTC_{m,t} \quad (32)$$

Since the constraints of the short-term model are mostly consistent with those of the long-term model, the following discussion will focus only on the constraints used in the short-term phase.

3.2.2. Knowledge-assisted operating constraint

In short-term optimization, it is necessary to establish selection rules for receiving and delivering oil tanks based on empirical knowledge.

The specific rules are as follows:

① Rules for Selecting Oil Receiving Tanks:

- Give priority to unloading oil into empty tanks.
- If condition a) is not satisfied, use the similarity calculation formula defined in Eq. (33) to determine the similarity between the crude oil to be unloaded and the residue oil type produced by the bottom oil in the tank. Select tanks with high similarity as the receiving tanks.

$$R_{v,c,k} = \left| \bigcap_{c \in \{UJ_v, BJ_k\}} CP_c \right|, \forall v, k \quad (33)$$

- Among the tanks with the highest similarity, choose the one with the largest capacity for unloading (Zhang et al., 2024).

② Rules for Selecting Oil Delivery Tanks:

Considering that selecting oil tanks capable of producing fewer types of residue oil in the early stages of scheduling can ensure a larger feasible domain in the later stages, the selection probability for oil delivery tanks is defined as follows:

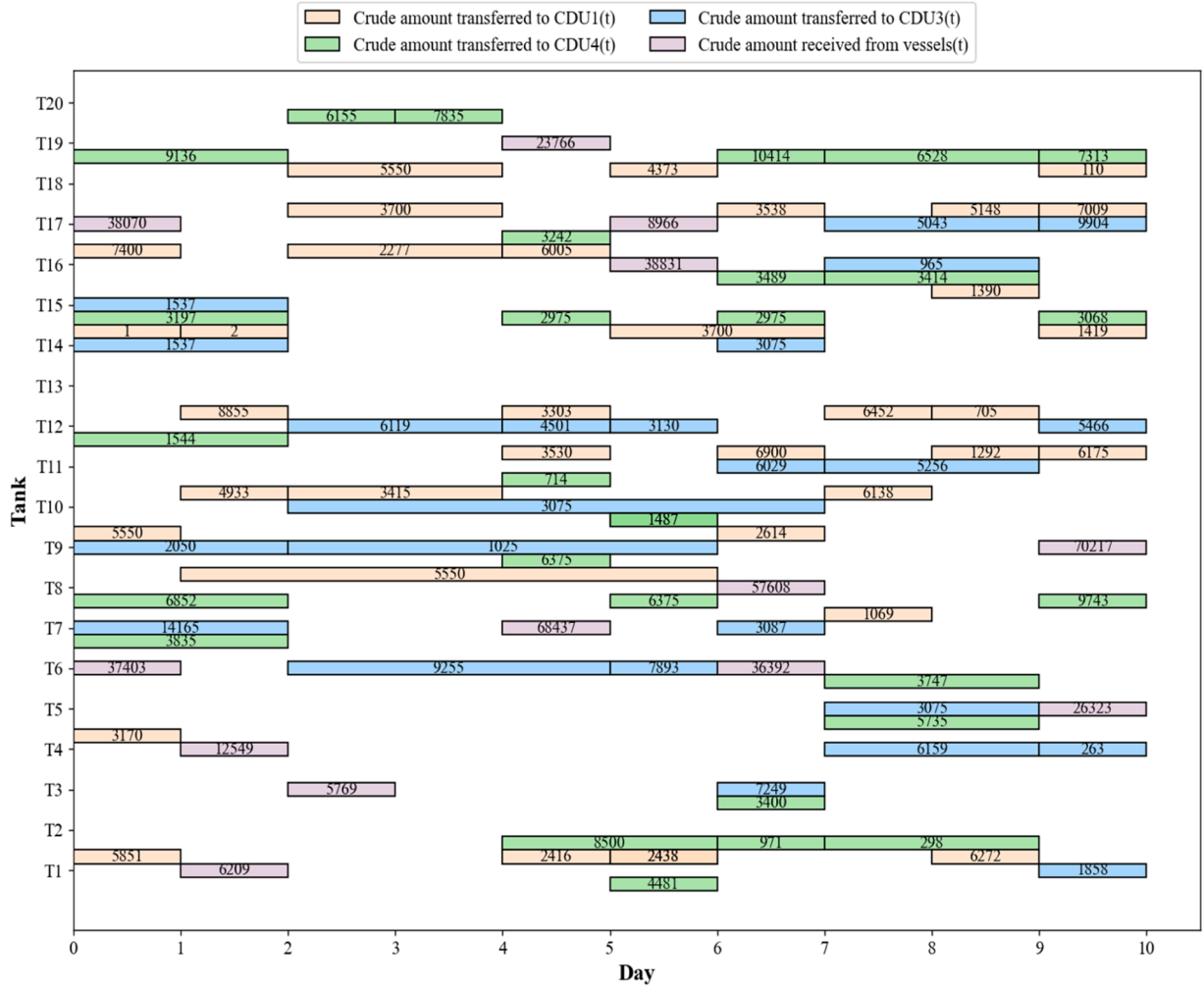


Fig. 14. Short-period scheduling scheme Gantt chart (Manual/PSO).

$$Prob_k = \frac{1}{\left| \bigcap_{c \in BJ_k} CP_c \right|} / \sum_{k \in AT} \frac{1}{\left| \bigcap_{c \in BJ_k} CP_c \right|}, k \in \text{tank} \quad (34)$$

$$X_{m,p,t} = \text{rand}, p \in MT_m, \forall m, t \quad (35)$$

$$CF_{m,p,t} = PL_m^{\min} + (PL_m^{\max} - PL_m^{\min}) \times \frac{X_{m,p,t}}{\sum_{p \in MT_m} X_{m,p,t}}, p \in MT_m, \forall m, t \quad (36)$$

In Eq. (34), the term "tank" represents the collection of oil tanks that meet the inventory lower limit requirement. Based on Eqs. (35) and (36), the initialization of oil delivery amounts is performed.

③ Determine the Set of Oil Delivery Tanks:

Based on the defined rules for selecting oil delivery tanks and the initial values for delivery volumes, identify the set of tanks to be used for oil delivery.

④ Determine the Oil Receiving Tanks:

Based on the rules for selecting oil receiving tanks, identify the tanks to be used for receiving oil. Calculate the amount of oil to be unloaded based on the capacity of the selected receiving tanks and the unloading rate.

3.2.3. Integration of long and short cycles

In practical applications, the crude oil scheduling problem is usually divided into two levels: long-term planning and short-term scheduling (Chunpeng and Gang, 2009). Long-term planning typically considers a

time scale of several weeks to several months, aiming to determine the input-output quantities and product ratios of each processing unit to meet market demand and production cost requirements. On the other hand, short-term scheduling typically considers a time scale of several days to several weeks, aiming to further optimize the oil receipts and deliveries of each tank based on the long-term plan to ensure the stability and efficiency of production operations. Therefore, after the optimization of long-term planning is completed, the optimization results can be used as the initial conditions for short-term scheduling. This can improve the efficiency and stability of short-term scheduling and better meet production demands. Specifically, the long-term planning model can be represented as follows:

$$(T, F, P) = \text{argmin} L(T, F, P) \quad (37)$$

In this formula, L represents the objective function of long-term planning, T represents the batch start-stop scheme, F represents the crude oil formulation scheme, and P represents the oil receipt and delivery scheme for each constant pressure device. This optimization problem can be solved using mathematical programming or optimization algorithms. Then, the optimization results of the long-term planning model can be used as the initial conditions for the short-term scheduling model, which can be represented as follows:

$$Q = f(T, F, P) \quad (38)$$

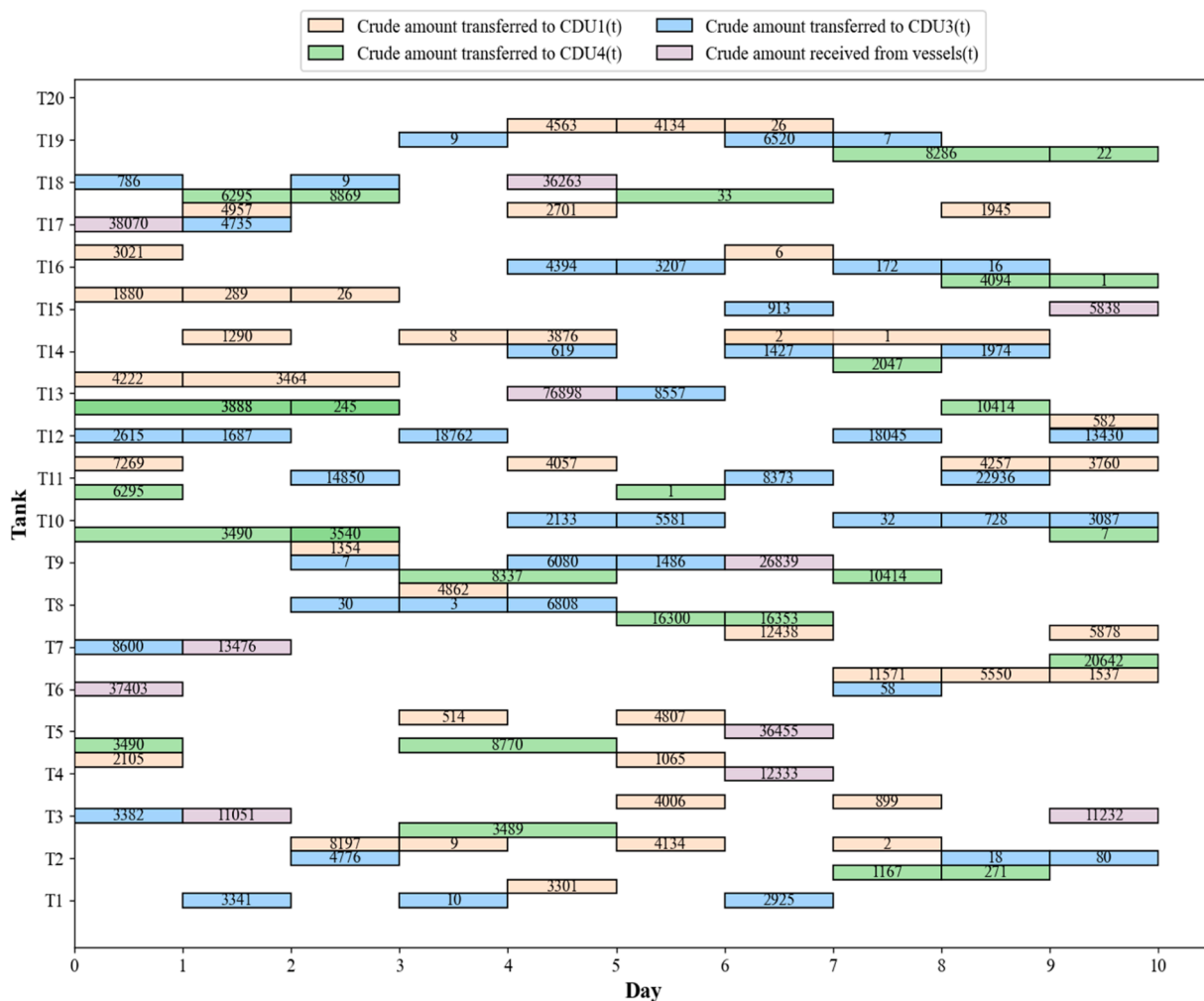


Fig. 15. Short-period scheduling scheme Gantt chart (PSO).

3.3. Model solving strategy

This article employs a hybrid optimization algorithm, MP/PSO, to solve the integrated crude oil scheduling optimization model for both long and short periods. The specific implementation steps are as follows:

Step1: Initialization: Initialize the population of solutions randomly or based on some heuristic method.

Step2: Long-term Planning Optimization: Implement the MP approach to optimize the long-term planning model, considering factors like batch start-stop schemes, crude oil formulation schemes, and oil receipt and delivery schemes for each constant pressure device.

Step3: Update the population of solutions with the optimized results from the long-term planning.

Step4: Short-term Scheduling Optimization: Utilize the PSO algorithm to further optimize the short-term scheduling model using the initial conditions obtained from the long-term planning phase. Consider factors such as minimizing cost, maximizing efficiency, and ensuring production stability in the short-term scheduling.

Step5: Termination Criteria: Define stopping criteria, such as reaching a maximum number of iterations or achieving a satisfactory solution quality.

Step6: Result Analysis and Post-processing: Evaluate the optimized solution obtained from the MP/PSO algorithm, considering various performance metrics such as production cost, stability, and efficiency.

The algorithm flowchart is depicted in the Fig. 5, and the solution process flowchart illustrating the information flow between long-term and short-term scheduling is shown in Fig. 6.

4. Industrial case validation

This paper uses two cases to verify the effectiveness of the proposed method. **Case 2** represents a real-world refinery plan from a coastal refinery, while **Case 1** is a simplified version of **Case 2**, primarily aimed at explaining the information flow between long-term and short-term scheduling. In **Case 2**, a detailed comparison was made among three different optimization methods: the MP/PSO method, which combines mathematical programming with PSO; the standalone PSO algorithm; and the PSO algorithm that uses a manually scheduled long-term plan as the initial condition (Manual/PSO). These comparative analyses provide a more comprehensive demonstration of the differences in optimization performance across the different methods.

The following are the shared data points for both cases:

A total of 12 types of crude oil are involved in the scheduling, with data on sulfur content, density, acid value, and types of produced residual oil shown in Table 1. The three atmospheric distillation units are CDU1, CDU3, and CDU4, with five side products: naphtha, straight-run gasoline, diesel, wax oil, and washing oil. The yield information for the 12 types of crude oil for each side product is presented in Table 2. The initial inventory, upper and lower limits, digestion limits, fixed output, and fixed consumption data for the wax oil tanks (containing W1 sulfur wax oil and W2 low sulfur wax oil) and the residue oil tanks (containing R1 coking material, R2 asphalt material, R3 catalytic material, and R4 residue additive) are detailed in Table 3.

The constraints for atmospheric distillation primarily include properties, side product yields, and processing loads, with specific data

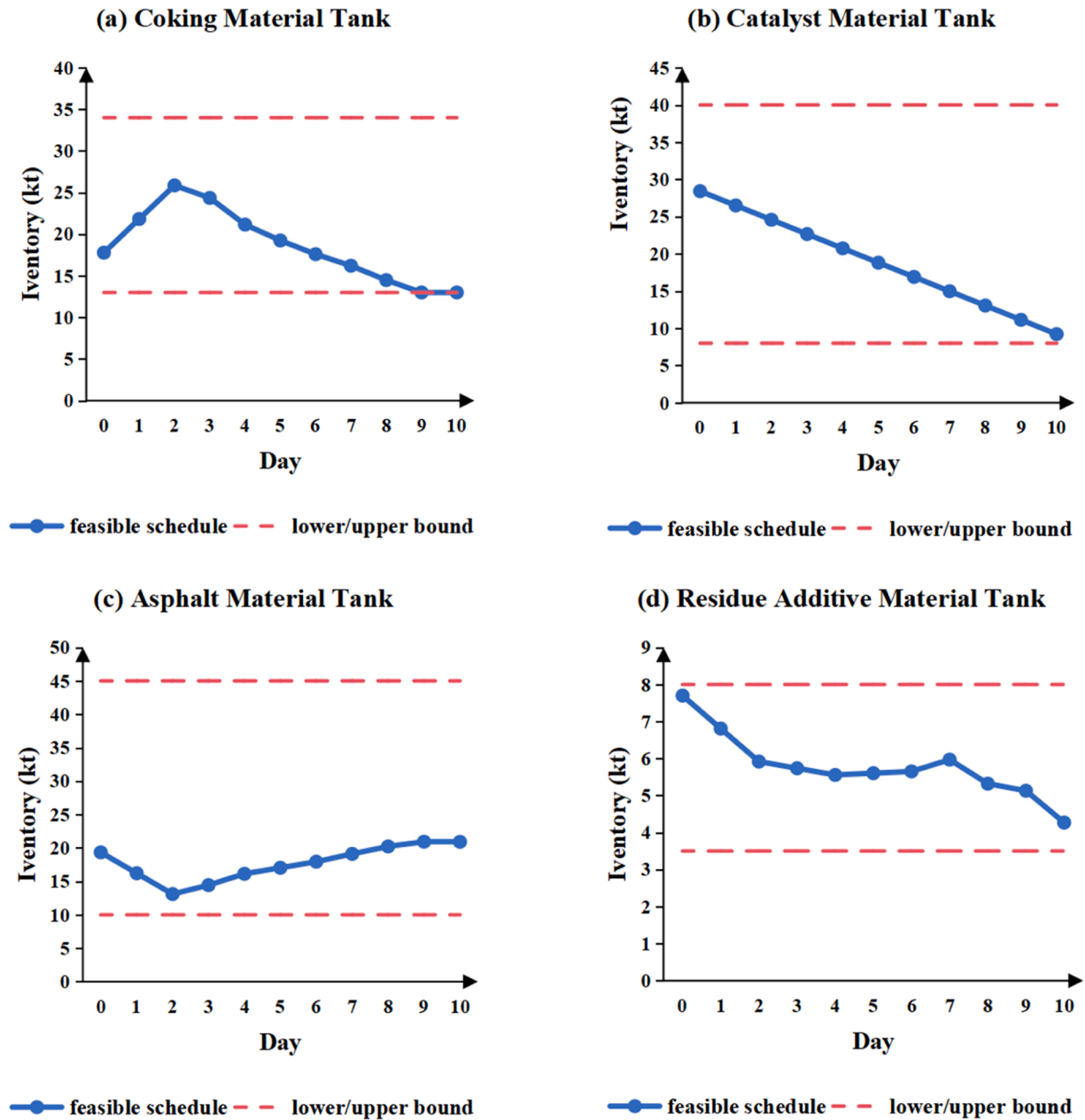


Fig. 16. Residue oil inventory change chart of MP/PSO. (a), (b), (c), and (d) represent coking material, catalyst material, asphalt material, and residue additive material, respectively.

provided in Table 4.

In this study, the variable scale of the MP is the product of the number of crude oil types $\times$ the number of tanks $\times$ the number of periods, the number of tanks $\times$ the number of CDUs $\times$ the number of periods, and the number of CDUs $\times$ the number of residual oil types $\times$ the number of periods. Clearly, as the scheduling scope increases, the scale of long-cycle scheduling optimization becomes substantial. The variable scale of the PSO is (maximum number of unloading tanks in one period + maximum number of loading tanks in one period) $\times 2 \times$ number of CDUs $\times$ number of periods. In Case 1, the maximum number of unloading and loading tanks in one period is 2, while in Case 2 it is 5.

Compared to standalone meta-heuristic algorithms (e.g., PSO), the proposed MP/PSO method in this article will require significantly more resource calls and time. The justification for this trade-off will be discussed in detail in the Case 2 section. Table 5 lists the problem sizes of Case 1 and Case 2, as well as the computational resources required under different methods. The CPU time of MP/PSO are the sum of the

CPU time of MP and PSO.

4.1. Case 1: simplified refinery plan to illustrate long-term and short-term scheduling interaction

This case uses a simple refining plan, with the relevant parameters and computation times shown in Tables 5 and 6.

The data for the crude oil tanker is shown in Table 7. The information for the crude oil storage tanks is provided in Table 8. Due to the complex composition of crude oil and the presence of a large amount of blended oil, the impact of physical tanks is considered to some extent in the resource-based long-term planning. The mixed oil in each tank is treated as a new crude oil for optimization purposes, and the components will be separated after the solution is completed. Additional information can be found in Tables 1, 2, 3, and 4.

The resource-oriented long-term scheduling results are shown in Fig. 7 and Fig. 8. Fig. 7 illustrates the residue oil output from the CDU

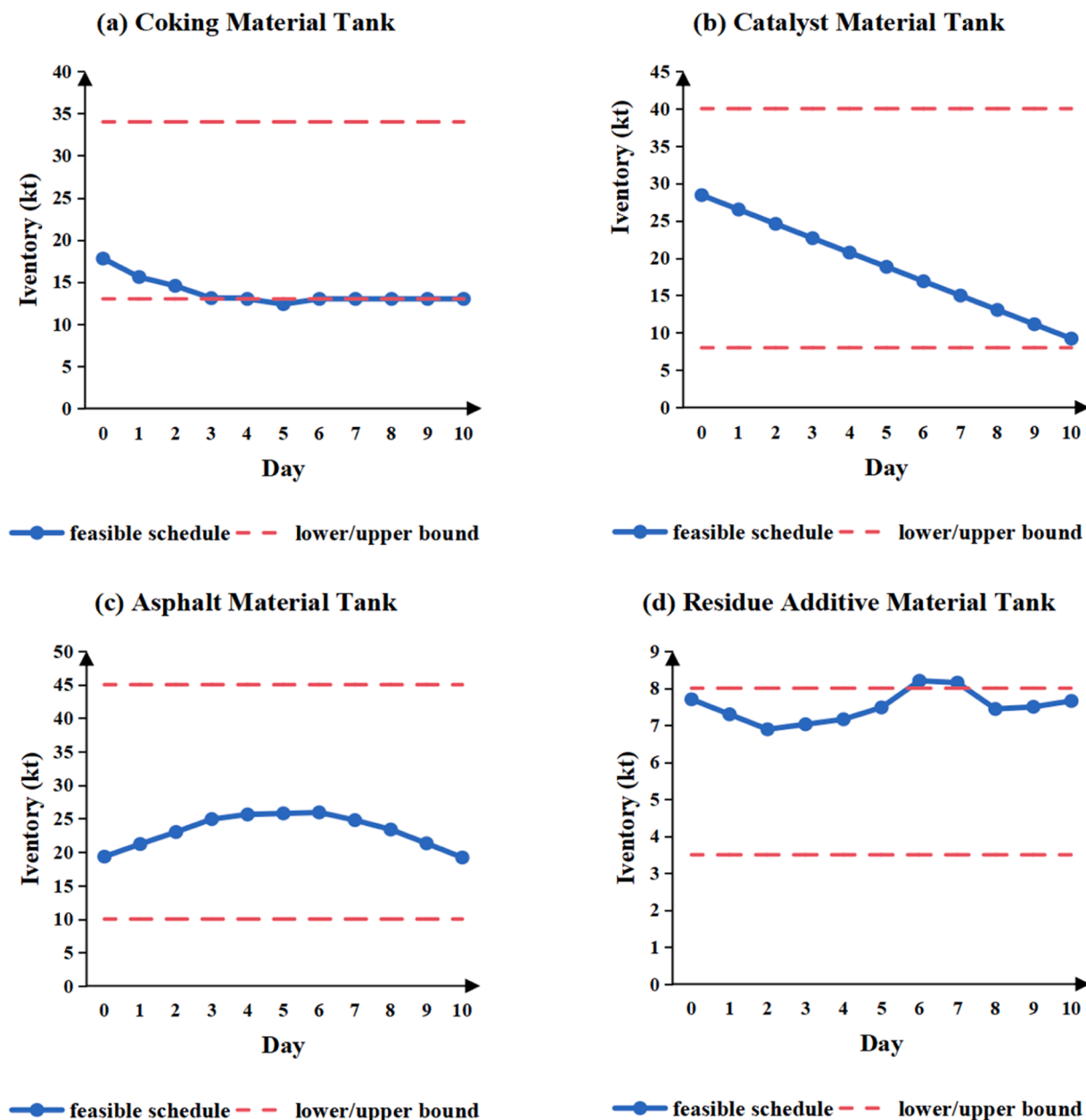


Fig. 17. Residue oil inventory change chart of Manual/PSO. (a), (b), (c), and (d) represent coking material, catalyst material, asphalt material, and residue additive material, respectively.

during the planning period, as well as the inflow and outflow status of various virtual tanks. Fig. 8 presents the oil blend formulations used by the CDU throughout the planning period. Fig. 9 contains the short-term scheduling results. As mentioned earlier, the short-term scheduling optimizes each cycle based on the long-term results, leading to minimal deviation from the long-term plan.

Taking CDU1 as an example, Fig. 8 indicates that in the first cycle, CDU1 used a blend of oils C1, C5, and C3. This result serves as the initial value for the short-term optimization, guiding the selection of the outflow tanks. As shown in Fig. 9, T10 and T12 were ultimately selected as the outflow tanks for CDU1 in the first cycle. Observing Table 4, we see that the initial oils in tanks T10 and T12 are C3 and C1, respectively, meeting the formulation requirements. It's important to note that to ensure the feasibility of the results, short-term optimizations may occasionally deviate from the long-term blend formulations (e.g., if there's insufficient stock of a specific oil), but they will not diverge from the residue oil outputs planned for each CDU in Fig. 7.

4.2. Case 2: Comparative study of a coastal refinery scheduling plan based on different algorithms

4.2.1. Initial parameter acquisition

This case is an expanded version of Case 1, with the relevant parameters and computation times shown in Tables 5 and 6. The data for the oil tankers is shown in Table 9, and the information for the crude oil storage tanks is provided in Table 10. Additional information can be found in Tables 1, 2, 3, and 4.

4.2.2. Comparison with Manual/PSO and the standalone PSO

Based on the initial parameters and constraints mentioned above, the long-term model was optimized using the GAMS solver, with an optimization time limit (reslim) set to 2000 s, an optimality criterion (optcr) of 0.01, and the MIP solver as CPLEX. The optimization results are illustrated in the Gantt chart shown in Fig. 10. The results indicate that CDU1 produces coking material for the first 2 days and asphalt material

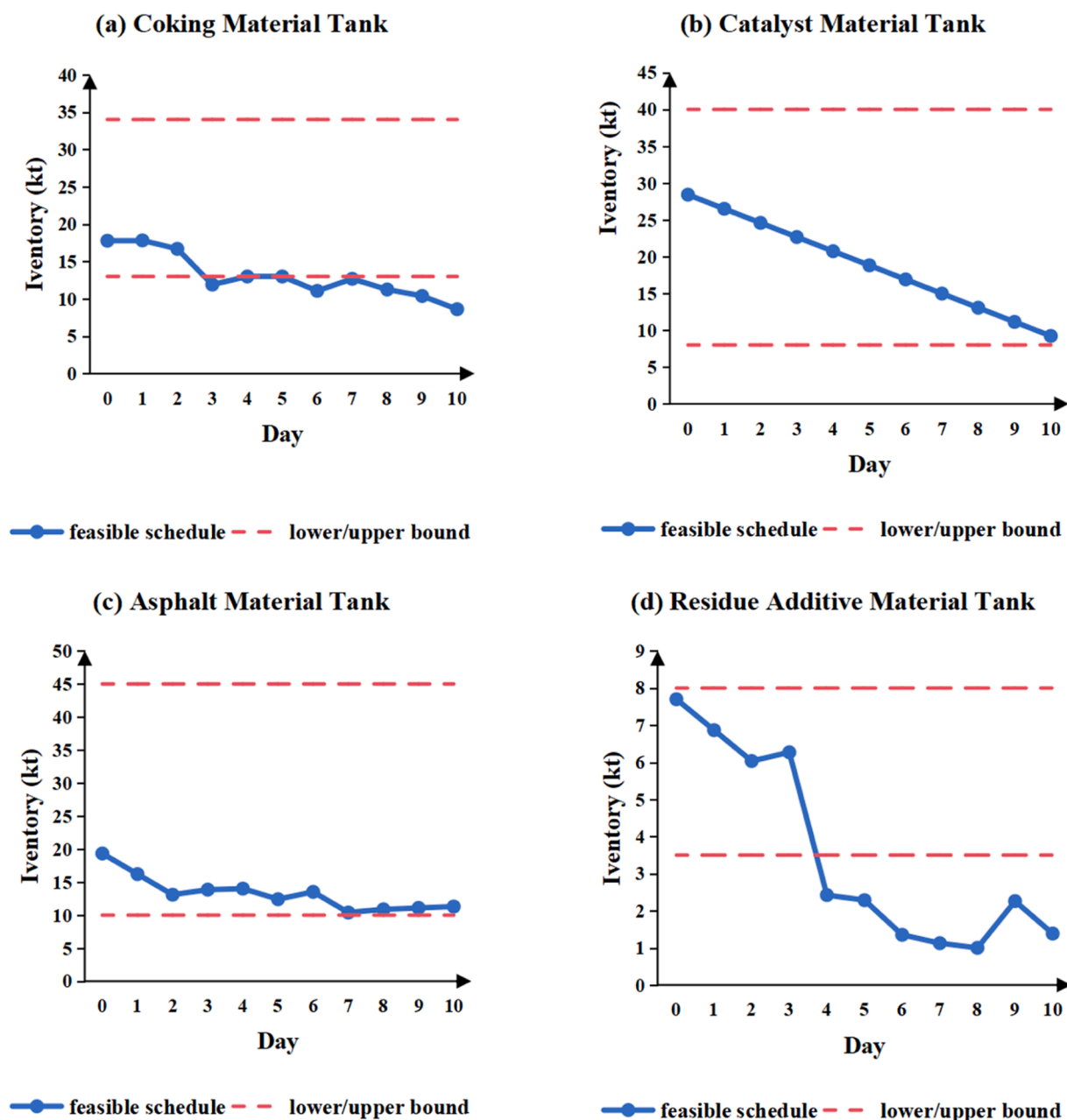


Fig. 18. Residue oil inventory change chart of PSO. (a), (b), (c), and (d) represent coking material, catalyst material, asphalt material, and residue additive material, respectively.

for the following 8 days; CDU3 produces residue additive material throughout the 10 days; and CDU4 produces coking material for the entire 10 days. The inventory changes of the virtual tanks for each oil type are shown in the chart, with yellow representing oil inflow and blue representing oil outflow. Fig. 11 depicts the proportion of required blend oils processed by each CDU during the optimized 10 cycles.

In order to demonstrate the validity of the MP/PSO framework, this study compares it with the Manual/PSO algorithm, which uses a manually scheduled long-term plan as the initial condition, and with the standalone PSO algorithm. The specific experimental methodology is as follows: for both the MP/PSO and Manual/PSO, the long-term scheduling plan is used as the initial parameter for short-term optimization. The short-term scheduling model is optimized using MATLAB. The length of each sub-cycle is set to 24 h, the maximum number of evaluations is set to 500, and optimization is repeated 20 times. For the standalone PSO algorithm, no long-term scheduling plan is used as an

initial parameter; instead, the initial schedule is randomly generated based on the initial inventory of each type of crude oil and the residual oil it can produce. The other optimization settings are the same as for the MP/PSO and Manual/PSO algorithms.

As shown in Fig. 12, during the 20 rounds of optimization, the Manual/PSO scheme had the fewest switching times, followed by MP/PSO, with the standalone PSO scheme performing the worst. This phenomenon can be attributed to the differing objectives and characteristics of the long-term plans. In the Manual/PSO scheme, the primary goal of manual scheduling is to maximize economic benefits, leading to relatively better performance. In contrast, the MP/PSO scheme not only aims to maintain high economic benefits but also ensures the feasibility of the solution, which leads to slightly lower economic benefits compared to the Manual/PSO scheme. As for the PSO, due to the absence of effective guidance from a long-term plan, it is more likely to fall into local optima, resulting in poorer performance compared to the other two schemes.

According to the results in Table 11, the constraint violation rate of the MP/PSO scheme is the lowest, followed by Manual/PSO, with PSO performing the worst. This result can also be attributed to the differences in the long-term plans. In the Manual/PSO scheme, although the initial conditions meet the constraints at the beginning of the cycle, the lack of comprehensive optimization over the entire cycle gradually reveals and amplifies the irrationality of the initial crude oil mix, leading to increasing constraint violations over time. In the PSO scheme, since the initial conditions are randomly set, the constraint violation rate is relatively higher. In contrast, the MP/PSO scheme takes into account the constraints of all cycles during the long-term optimization phase, resulting in a more reasonable and reliable initial resource scheduling, which lays a solid foundation for subsequent short-term optimization. Table 12 summarizes the data from Fig. 12 and Table 11, where mean and std represent the average and standard deviation of switching times during the 20 rounds of optimization, respectively. The Feasible Rate (FR) indicates the probability of obtaining a feasible solution during the 20 rounds of optimization, i.e., the probability of no constraint violations.

Taking the optimization results from the 3rd round as an example, the MP/PSO scheme had 20 switches and 0 constraint violations, the Manual/PSO scheme had 18 switches and 3 constraint violations, while the standalone PSO scheme had 24 switches and 55 constraint violations. By comparing Fig. 13, Fig. 14, and Fig. 15, further insights into the scheduling characteristics of each scheme can be observed. In terms of switching frequency and scheduling continuity, both MP/PSO and Manual/PSO performed better, showing stronger continuity and relatively fewer switches. In contrast, the standalone PSO scheme exhibited higher discreteness, with frequent switches and significantly less continuity. Moreover, it is particularly noteworthy that only the MP/PSO scheme provided a feasible solution without any constraint violations. This indicates that the MP/PSO scheme better balances the global scheduling feasibility and constraint conditions during the optimization process, highlighting its superiority in addressing complex resource scheduling problems.

In this case, due to the significant constraint violations observed under the PSO method, we will focus on discussing some key instances of constraint violations. According to the results shown in Fig. 16, the MP/PSO method exhibited no constraint violations, demonstrating its strong feasibility. In contrast, Fig. 17 reveals that the Manual/PSO approach, constrained by the limitations of manual scheduling, faced supply shortages of certain oil types in the later stages due to the lack of comprehensive planning over the entire cycle. This led to violations of the constraints related to coking materials and Residue Additive Material. Furthermore, as shown in Fig. 18, the PSO method not only violated the constraints for catalytic materials and Residue Additive Material, but also showed no signs of recovery, indicating poor optimization performance.

In summary, although both MP/PSO and Manual/PSO methods introduce additional computational overhead due to long-term planning and scheduling, the optimization outcomes are significant. Their optimization results can be used as initial conditions for short-term scheduling, effectively improving the efficiency and stability of short-term scheduling and better meeting production demands. Compared to manual scheduling, the long-term scheduling approach based on mathematical programming demonstrates more comprehensive optimization capabilities. Moreover, as shown in Table 6, the computation time for the mathematical programming approach is significantly less than the time required for manual scheduling. This case demonstrates that the long-term scheduling method based on mathematical programming not only ensures the feasibility of the solution but also achieves a better balance between production constraints and economic benefits, showcasing superior overall scheduling performance. This further confirms the practical value of mathematical programming in solving complex production scheduling problems and its potential in enhancing production system performance.

5. Conclusion

In practice, resource-oriented long-term crude oil production plans may fail to meet the needs of decision-makers. Conversely, for practical operational scheduling, especially in large-scale scheduling, they typically require a combination of long-term production plans and short-term scheduling plans. Therefore, the goal of this paper is to integrate long and short-term cycles and make resource-oriented long-term plans easy to handle and decompose. In this study, we propose an integrated scheduling optimization method that combines knowledge and experience with long and short-term cycles. This method combines mathematical programming with particle swarm optimization and has successfully achieved experimental results on practical cases with large-scale characteristics.

In the experiments, we compared MP/PSO with the standalone PSO and Manual/PSO. The results showed that the standalone PSO, due to its randomness, led to numerous constraint violations and frequent equipment switches. Manual/PSO, benefiting from the long-term plan generated by manual scheduling, resulted in the fewest equipment switches but still violated a few constraints, making it an infeasible solution. Ultimately, the conclusion was that MP/PSO can produce a well-performing and feasible short-term schedule within an appropriate time frame.

In the future, we will further research scheduling optimization based on the combination of long and short-term cycles under an uncertain model to meet the growing demand for large-scale scheduling in coastal refineries. Additionally, we will study the modification of long-term plans based on violations of constraints in short-term scheduling scenarios to achieve closed-loop planning to scheduling.

CRediT authorship contribution statement

Renchu He: Conceptualization, Methodology, Software, Resources, Formal analysis, Investigation, Writing – review & editing. **Yunhao Xie:** Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing. **Shiwei Zhang:** Methodology, Visualization, Resources, Data curation, Investigation. **Feng Xu:** Methodology, Software, Conceptualization, Investigation. **Jian Long:** Software, Conceptualization, Funding acquisition, Methodology, Investigation, Visualization.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Data availability

The data used has already been indicated in the original manuscript

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